

APPENDIX 2:

SOFTWARE DESIGN DOCUMENT (SDD)		
PRESENTED BY: AuthenticityHub	PRESENTED TO: PM. TS. DR. Rohani Binti Abu Bakar	DATE: 15/06/2025

TEAM MEMBERS

Chief Executive Officer  Qamhan Khaled Mohammed Abdo CB23009	Chief Business Analyst  Abdullah Mueataz Adel Ahmed CB23001	Chief Developer  Banabila Salman Mohammed CB23142	Software Quality Assurance  AL-QARMWSHI ABDULLAH FAEZ ABDULLAH CB23005
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Submitted to the Software Design Workshop (BCS2343).

Bachelor of Computer Science (Software Engineering) with Honours.

Faculty of Computing, Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA).

DOCUMENT APPROVAL

	Name	Date
Authenticated by:  Project Leader	Qamhan Khaled Mohammed Abdo	
Approved by:  Project Manager	Abdullah Mueataz Adel Ahmed	
Approved by: Client	PM. TS. DR. Rohani Binti Abu Bakar	

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LIST OF ABBREVIATIONS

Software Design Document	SDD
Entity Relationship Diagram	ERD
Primary Key	PK
Foreign Key	FK
Malaysian Communications and Multimedia Commission	MCMC
Model-View-Controller	MVC

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Design Document (SDD) is to provide a comprehensive design blueprint for the MySebenarnya System, commissioned by the Malaysian Communications and Multimedia Commission (MCMC). This document outlines the architectural structure, key components, and operational workflow required to build and maintain the system.

This Software Design Document (SDD) serves as a foundational reference for the development of the Inquiry System, a digital solution designed to streamline the management of public inquiries. The document is intended for a wide audience including system developers, designers, stakeholders, administrators, and testers. It provides a structured overview of the system's architecture, component interactions, and design rationale to ensure all parties involved have a clear understanding of the project's goals and implementation pathway.

This SDD has been prepared in alignment with best practices in software engineering and reflects the collaborative efforts of the project team to build a reliable, user-friendly, and transparent inquiry management platform. Each module outlined in this document plays a vital role in the seamless operation of the system, from user registration to inquiry tracking and resolution.

By following the design outlined in this document, the team aims to ensure the system is scalable, maintainable, and capable of meeting both current and future needs of the intended users and administrative bodies.

1.2 System Identification

This Software Design Document (SDD) presents the system architecture and design specifications for the AuthenticityHub System, an initiative by AuthenticityHub to modernize the way public inquiries are managed. The system empowers users to create and manage their accounts, submit official inquiry forms, and ensure those inquiries are directed to the correct agencies or authorities for further action.

Through real-time progress tracking and structured workflows, AuthenticityHub aims to enhance clarity, build trust between the public and service providers, and ensure timely responses to pressing public matters. The system serves as a centralized platform focused on improving service efficiency, communication flow, and public accountability.

System Title	MySebenarnya
System Abbreviation	AH
Year of Development	2025
System Version	1.0 (V1.0)
System Document ID Format	<System abbreviation><Subsystem abbreviation><Year of development><System Version>
System Document ID	SDD-AH-2025-V1.0

Subsystem Title	Public Inquiry Management Subsystem
Subsystem Abbreviation	PIMS
Subsystem Document ID	AH-PIMS-2025-V1.0
Requirement ID Format	<Document System Abbreviation><Requirement abbreviation><Requirement number>
Requirement ID	SRS-REQ-100
Design ID	SDD-REQ-100
Document System Name	Software Design Document
Document System Abbreviation	SDD
Package ID	AH-PG-100
Class ID	AH-PG-100-CL-101

1.3 System Overview

1.3.1: Manage User (User Registration, Login and Profile Management): This module is the gateway for users to access the AuthenticityHub platform. It allows users to create accounts, log in securely, and manage personal profiles such as email, password, and contact information. The system maintains authentication control and ensures that only legitimate users can submit inquiries. The public user can register a new account, log in with their credentials, and update their profile details at any time, establishing a secure and personalized experience for each user interacting with the system. MCMC acts as the system administrator, monitoring user registration activities and handling security-related functions such as resetting passwords or deactivating suspicious accounts. Agencies do not have direct involvement in this module, but user information may be displayed to them when handling assigned inquiries.

1.3.2: Manage Inquiry Form Submission: This module provides a structured digital form that users can use to submit inquiries, reports, or complaints. It captures detailed information related to the user's issue and may include attachments like photos or documents to support their claim. The public user accesses the inquiry form, fills in the required details, selects the issue category, and submits it through the platform. This enables them to raise issues in an official and trackable way. MCMC reviews submitted forms for completeness and relevance, verifies the accuracy of the information, and decides whether the inquiry is valid and ready to be forwarded for assignment. Agencies do not interact with this module directly but will receive the inquiry only after MCMC has approved and assigned it.

1.3.3: Inquiry Assignment (to Agencies): This module enables MCMC to assign submitted inquiries to the appropriate agency based on the nature, category, and scope of the issue. The goal is to ensure that each inquiry reaches the right party for resolution. The public user is not involved in the assignment process but will be notified once their inquiry is forwarded to a relevant agency for action. MCMC evaluates each verified inquiry and uses this module to assign it to the correct agency, typically based on issue type, location, or the agency's area of responsibility. The agency receives the assigned inquiry along with all supporting data and is expected to acknowledge the assignment and begin working on the resolution.

1.3.4: Inquiry Progress Tracking: This module tracks the lifecycle of each inquiry, from submission to resolution. It ensures transparency by updating the status and making it visible to all relevant parties, including status labels, timestamps, and action logs. The public user can log in to view the status of their inquiry, check progress updates, and read any agency responses, helping build trust by keeping the user informed. MCMC monitors the progress of all inquiries, ensures agencies are updating statuses, and may intervene if inquiries are delayed or left unresolved. Agencies use this module to update the status of each inquiry as they work on it, marking it as "In Progress", "Resolved", or "Closed", and may add comments or feedback for the user.

1.4 References

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2. DATA DESIGN

2.1 ENTITY RELATIONSHIP DIAGRAM (ERD)

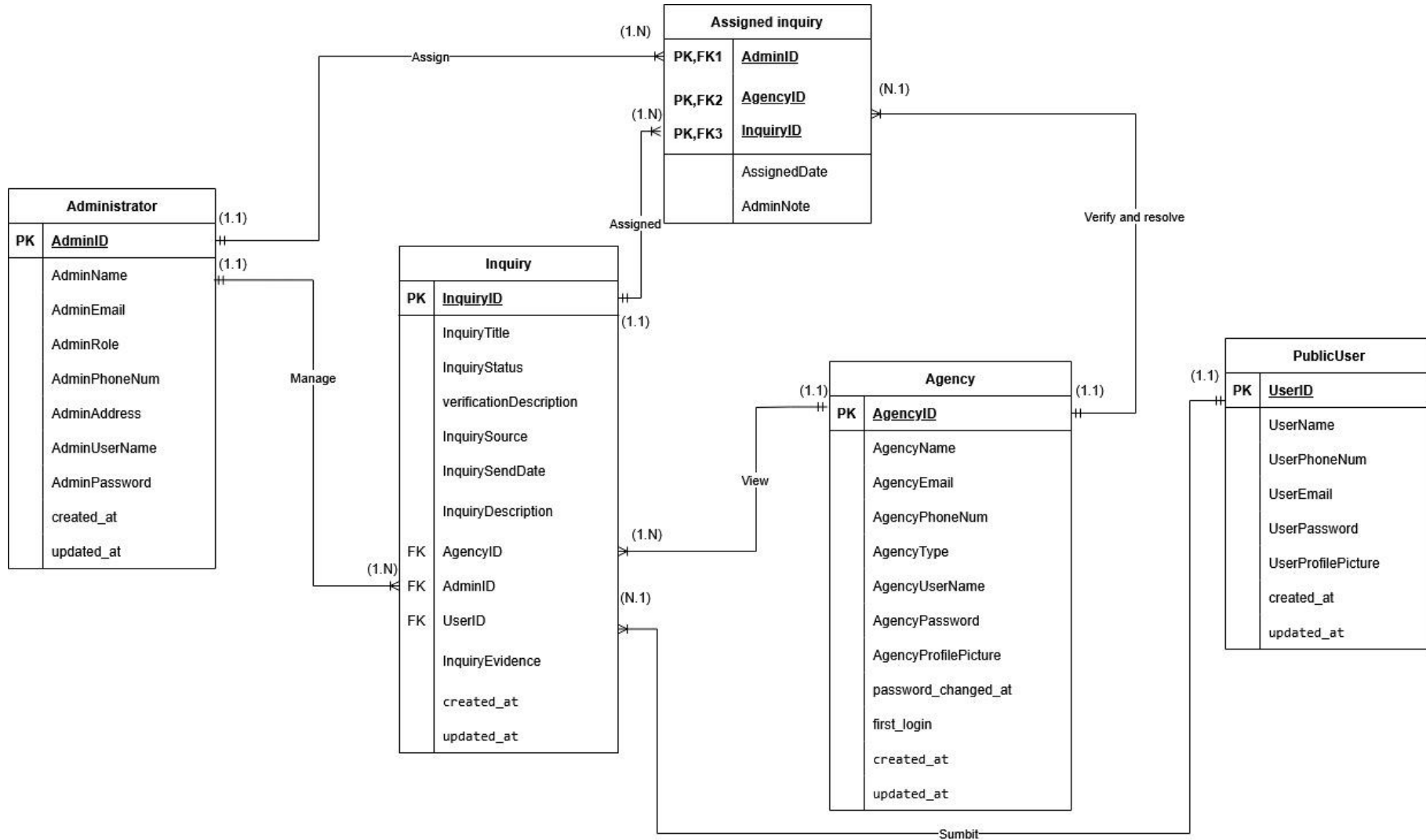


Figure 2.1 Entity Relationship Diagram (ERD) for AuthenticityHub System

The Entity-Relationship Diagram (ERD) illustrates a system designed to manage public inquiries and their resolution through administrative assignment and agency action. This diagram includes seven main entities: Administrator, PublicUser, Agency, Inquiry, AssignedInquiry, Report, and Inquiry, highlighting their relationships and attributes.

1. Administrator Entity:

This table stores information about system administrators who manage the inquiry system. It includes basic details like **AdminID**, **AdminName**, **AdminEmail**, **AdminRole**, and login credentials. Administrators are responsible for assigning inquiries to the right agencies and overseeing the resolution process.

2. PublicUser Entity:

This table contains information about citizens who submit inquiries to the system. It includes **UserID**, **UserName**, **UserEmail**, **UserPhoneNum**, and **UserProfilePicture**. Public users create inquiries when they have questions or complaints about communications and multimedia services.

3. Agency Entity:

This table stores details about government agencies that handle different types of inquiries. It includes **AgencyID**, **AgencyName**, **AgencyEmail**, **AgencyType**, and login credentials. Each agency specializes in resolving specific types of inquiries based on their expertise.

4. Inquiry Entity:

This is the main table that stores all inquiry details submitted by public users. It includes **InquiryID**, **InquiryTitle**, **InquiryDescription**, **InquiryStatus**, **InquiryEvidence**, and links to the user who submitted it. This table tracks each inquiry from submission to resolution.

5. AssignedInquiry Entity:

This table manages the assignment of inquiries to agencies by administrators. It connects **AdminID**, **AgencyID**, and **InquiryID** with an **AssignedDate** and **AdminNote**. This ensures each inquiry is properly assigned and tracked.

Relationships:

- 1. Public users submit inquiries through the system
- 2. Administrators review and assign inquiries to appropriate agencies
- 3. Agencies work on resolving the assigned inquiries
- 4. The system tracks all activities and maintains records for accountability

2.2 DATA DICTIONARY

2.2.1 Table Name : Administrator

Field Name	Description	Data Type	Constraint
AdminID	Administrator unique ID	BIGINT UNSIGNED	PK, AUTO_INCREMENT, NOT NULL
AdminName	Administrator full name	VARCHAR(255)	NOT NULL
AdminEmail	Email address for login	VARCHAR(255)	UNIQUE, NOT NULL
AdminRole	Role of administrator	VARCHAR(100)	NOT NULL
AdminPhoneNum	Administrator phone number	VARCHAR(20)	NOT NULL
AdminAddress	Administrator address	VARCHAR(255)	NOT NULL
AdminUserName	Username for login	VARCHAR(100)	UNIQUE, NOT NULL
AdminPassword	Hashed password for login	VARCHAR(255)	NOT NULL
created_at	Record creation timestamp	TIMESTAMP	NOT NULL
updated_at	Record last update timestamp	TIMESTAMP	NOT NULL

2.2.2 Table Name : PublicUser

Field Name	Description	Data Type	Constraint
UserID	Public user unique ID	BIGINT UNSIGNED	PK, AUTO_INCREMENT, NOT NULL
UserName	User full name	VARCHAR(255)	NOT NULL
UserPhoneNum	Contact phone number	VARCHAR(20)	NULLABLE
UserEmail	Email address for login	VARCHAR(255)	UNIQUE, NOT NULL
UserPassword	Hashed password for login	VARCHAR(255)	NOT NULL
UserProfilePicture	Profile picture file path	VARCHAR(255)	NULLABLE
created_at	Record creation timestamp	TIMESTAMP	NOT NULL
updated_at	Record last update timestamp	TIMESTAMP	NOT NULL

2.2.3 Table Name : Agency

Field Name	Description	Data Type	Constraint
AgencyID	Agency unique ID	BIGINT UNSIGNED	PK, AUTO INCREMENT, NOT NULL
AgencyName	Official name of the agency	VARCHAR(255)	NOT NULL
AgencyEmail	Agency contact email	VARCHAR(255)	NOT NULL
AgencyPhoneNum	Agency contact phone number	VARCHAR(20)	NOT NULL
AgencyType	Type/category of agency	VARCHAR(100)	NOT NULL, DEFAULT('Default')
AgencyUserName	Username for login	VARCHAR(100)	UNIQUE, NOT NULL
AgencyPassword	Hashed password for login	VARCHAR(255)	NOT NULL
first_login	Flag for first login password change	TINYINT(1)	DEFAULT(1)
password_changed_at	Timestamp when password was last changed	TIMESTAMP	NULLABLE
AgencyProfilePicture	Agency profile picture file path	VARCHAR(255)	NULLABLE
created_at	Record creation timestamp	TIMESTAMP	NOT NULL
updated_at	Record last update timestamp	TIMESTAMP	NOT NULL

2.2.4 Table Name : AssignedInquiry

Field Name	Description	Data Type	Constraint
AdminID	Admin who assigned inquiry	BIGINT UNSIGNED	FK, NOT NULL
AgencyID	Agency assigned to inquiry	BIGINT UNSIGNED	FK, NOT NULL
InquiryID	Assigned inquiry ID	BIGINT UNSIGNED	FK, NOT NULL
AssignedDate	Date and time of assignment	TIMESTAMP	NOT NULL
AdminNote	Note from admin	TEXT	NULLABLE

2.2.5 Table Name : Inquiry

Field Name	Description	Data Type	Constraint
InquiryID	Inquiry unique ID	BIGINT UNSIGNED	PK, AUTO_INCREMENT, NOT NULL
InquiryTitle	Title of the inquiry	VARCHAR(255)	NOT NULL
InquiryStatus	Current inquiry status	VARCHAR(100)	NOT NULL
VerificationDescription	Verification notes and results	TEXT	NULLABLE
InquirySource	Source of the inquiry	VARCHAR(100)	NOT NULL
InquirySendDate	Date inquiry was submitted	TIMESTAMP	NOT NULL
InquiryDescription	Full description of inquiry	TEXT	NOT NULL
InquiryEvidence	Attached media/document file paths	TEXT	NULLABLE
AgencyID	Agency handling the inquiry	BIGINT UNSIGNED	FK, NULLABLE
AdminID	Admin managing the inquiry	BIGINT UNSIGNED	FK, NULLABLE
UserID	Public user who submitted inquiry	BIGINT UNSIGNED	FK, NOT NULL
created_at	Record creation timestamp	TIMESTAMP	NOT NULL
updated_at	Record last update timestamp	TIMESTAMP	NOT NULL

3. GENERAL ARCHITECTURE

3.1 Architectural Pattern: MVC

The system utilizes the design pattern known as **Model-View-Controller (MVC)**. The architecture of this system is composed of three distinct layers, which are as follows:

1. **Model:** Represents the application's core business logic and handles data operations, such as inquiry management, user records, and agency assignments.
2. **View:** Manages the user interface, allowing users such as administrators, public users, and agencies to interact with the system and view information relevant to their roles.
3. **Controller:** Acts as an intermediary between the View and Model, processing user inputs, triggering data operations, and determining appropriate views to render.

The MVC pattern's modular structure makes the system highly **maintainable** and **scalable**. Modifications to one component such as redesigning the user interface in the View or updating the database schema in the Model can be implemented with minimal impact on the other components. Furthermore, this pattern is well-suited for **object oriented programming**, enabling the use of encapsulation, inheritance, and polymorphism to streamline functionality and encourage code reuse. This makes MVC an ideal architectural choice for the **Public Inquiry Management System**, where multiple modules and user roles must operate seamlessly and efficiently within a robust, well-organized framework.

3.2. Component Overview

The "AuthenticityHub" system adopts a modular architecture, segmenting its core functionality into independent yet interconnected modules. Each module is designed to handle a specific part of the system, ensuring streamlined operations and smooth interaction between components. The key modules developed are Manage User Module, Manage Inquiry Form Submission Module, Inquiry Assignment Module, and Inquiry Progress Tracking Module

3.2.1 Module 1: Manage User

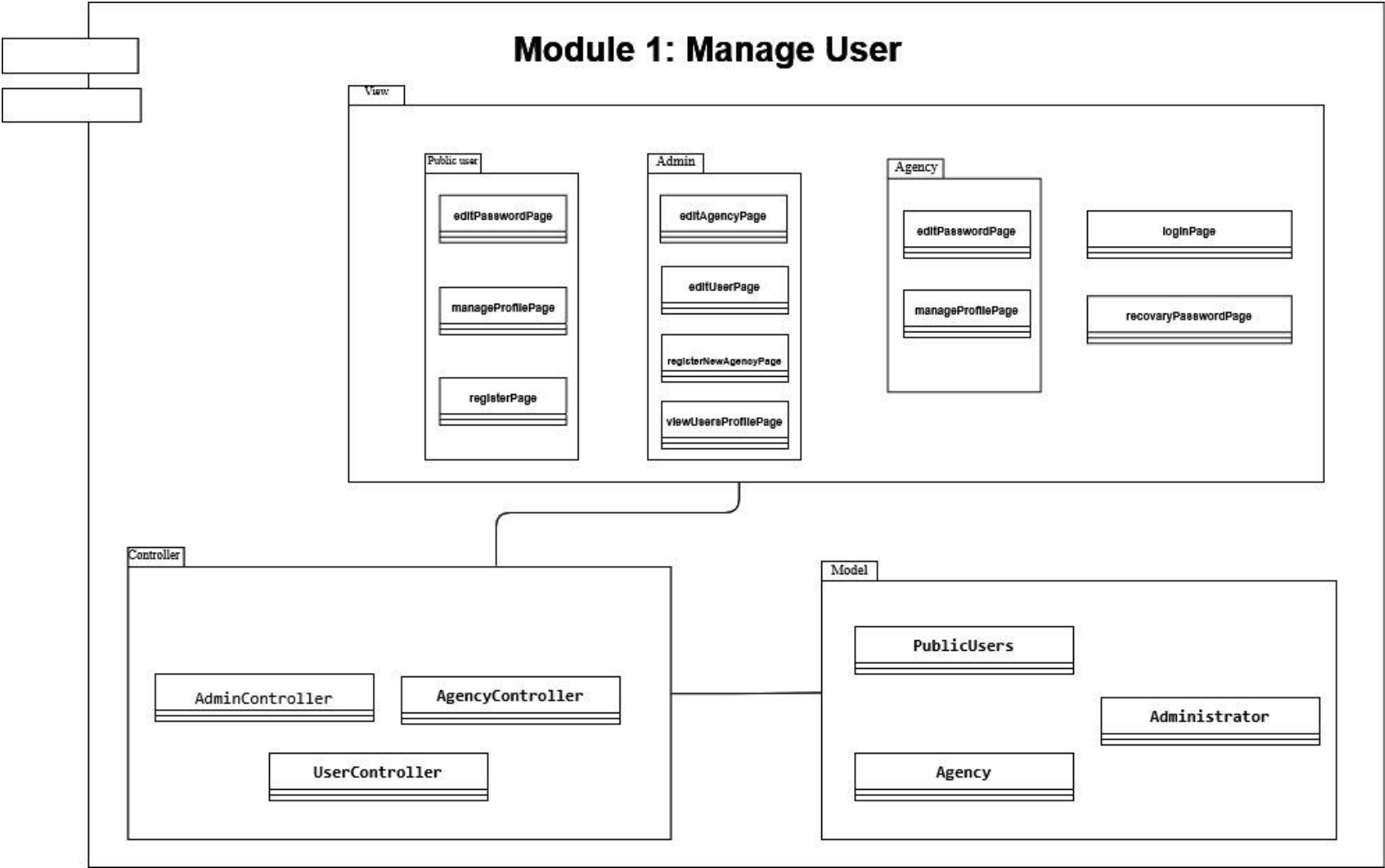


Figure 3.2.1 Component Module 1: Manage Use

3.2.1.1 View Package

Class Name	Description
RegisterPage	User registration form that allows new public users to create an account by providing their personal information and credentials.
ManageProfilePage (PublicUser)	Public User profile management interface that enables logged-in users to view, update, and manage their personal profile information.
ManageProfilePage (Agency)	Agency profile management interface that enables logged-in users to view, update, and manage their personal profile information.
EditPasswordPage (PublicUser)	Password change form that allows Public user to securely update their account password by providing current and new password credentials.
EditPasswordPage (Agency)	Password change form that allows Agency to securely update their account password by providing current and new password credentials.
RegisterNewAgencyPage	Admin-only form that allows administrators to register and add new agencies to the system with relevant agency details.
ViewUsersProfilePage	Admin dashboard that displays profile information of all registered users and agencies, providing administrators with an overview of system users.
editAgencyPage	Admin interface that allows administrators to modify and update existing agency information and details in the system.
editUserPage	Admin interface that enables administrators to edit and update public user account information and profile details.
loginPage	A login interface that provides authentication functionality for all user types (public users, MCMC staff, and agency staff) to access the system using
recoveryPasswordPage	A password recovery interface that allows users to reset their forgotten passwords by providing their email address or username for password reset

3.2.1.2 Controller Package

Class Name	Description
UserController	Handles user account management functionality including user registration, profile management, and password updates for public users.
AdminController	Manages administrative functions such as registering new agencies, viewing user and agency profiles, and editing user/agency information in the system.
AgencyController	Handles agency-specific operations and management of agency-related data and functionality within the system.

3.2.1.3 Model Package

Class Name	Description
PublicUsers	Represents individual user accounts in the system. Stores user profile information, authentication credentials, and manages user account data for
Agency	Represents agency entities in the system. Stores agency information, details, and manages agency-related data that can be registered and managed by
Administrator	Represents admin user accounts with elevated privileges. Manages administrator authentication and stores admin-specific data for users who can register agencies, view profiles, and edit user/agency information in the system.

3.2.2 Module 2: Manage Inquiry Form Submission

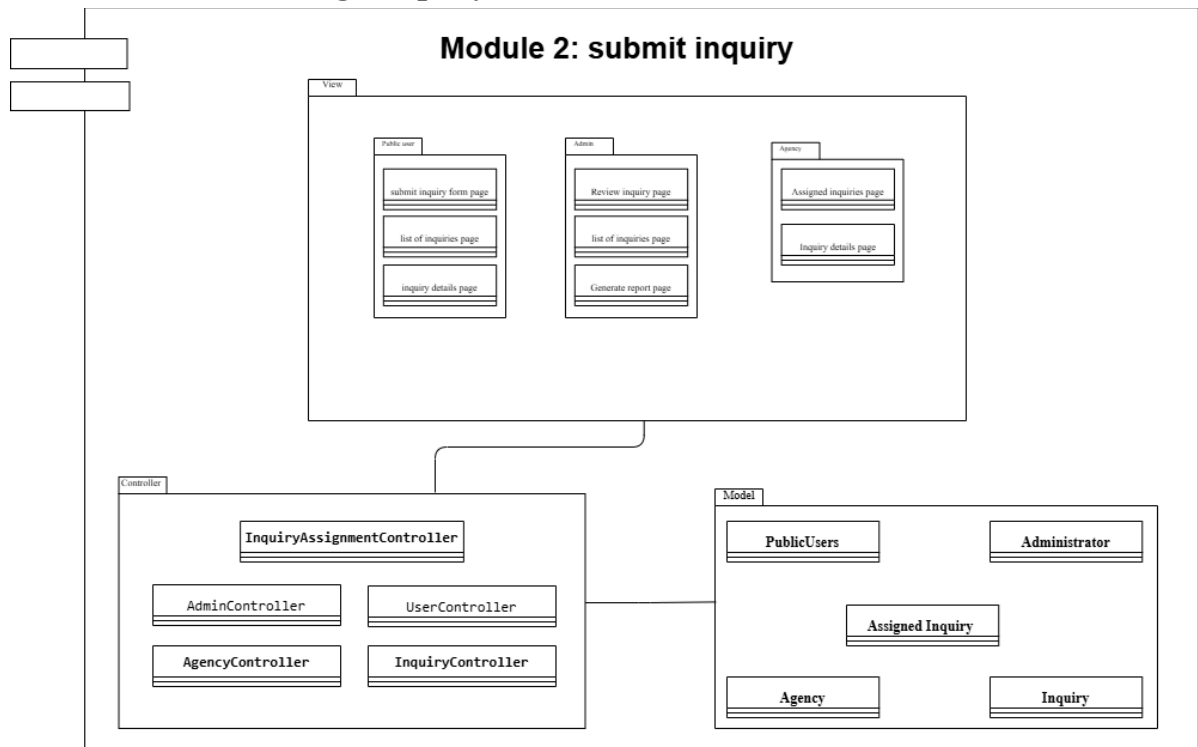


Figure 3.2.2 Component Module 2: Manage Inquiry Form Submission

3.2.2.1 View Package

Class Name	Description
Submit Inquiry Form Page	Interface for users to submit inquiries with evidence. Supports file uploads and input validation.
List of Inquiries Page	Displays user-submitted inquiries and others inquiries with statuses. Includes filters for tracking and reviewing submissions.
Inquiry Details Page	Shows full details and updates for an inquiry.
Review Inquiry Page	Enables admins to evaluate, flag, or assign inquiries. Displays submission details and evidence.
List of Inquiries Page	Shows newly submitted inquiries for review. Allows sorting and filtering.
Generate Report Page	Generates statistical and downloadable inquiry reports. Offers formats like PDF, Excel, and visual graphs.
Assigned Inquiries Page	Lists inquiries assigned to the agency. Allows status filtering and quick access to details.
Inquiry Details Page	Shows complete inquiry information for agencies. Includes evidence files and action buttons for acceptance/rejection.

3.2.2.2 Controller Package

Class Name	Description
InquiryAssignmentController	Manages the assignment of inquiries to agencies with bulk assignment capabilities and assignment tracking.
AdminController	Handles admin operations including inquiry review, assignment management, and report generation.
UserController	Manages public user operations including inquiry submission, viewing, authentication, and profile management.
AgencyController	Controls agency operations including dashboard, inquiry management, authentication, and status updates.
InquiryController	Manages inquiry operations including submission, tracking, status updates, and user inquiry management.

3.2.2.3 Model Package

Class Name	Description
Inquiry	Represents inquiries with submission details, evidence files, status tracking, and user relationships.
PublicUser	Represents public users who submit inquiries with authentication and profile management capabilities.
Agency	Represents agencies that process inquiries with authentication and inquiry management functionality.
Administrator	Represents administrators who manage the system with inquiry assignment and reporting capabilities.
AssignedInquiry	Represents the assignment relationship between inquiries and agencies with tracking information.

3.2.3. Module 3: Inquiry Assignment

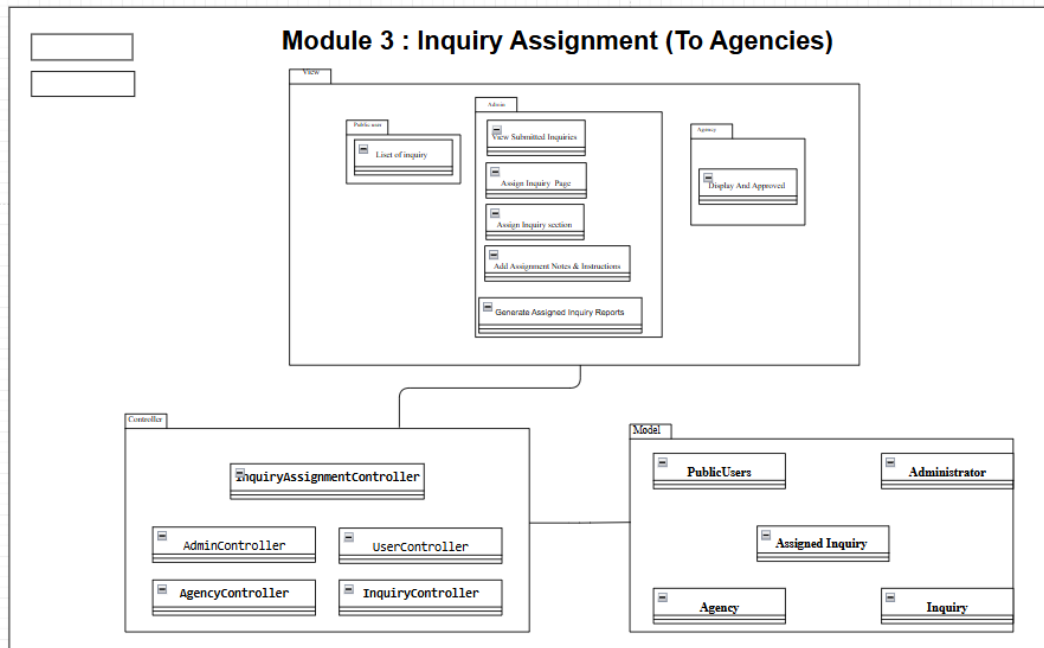


Figure 3.2.3 Component Module 3: Inquiry Assignment (to Agencies)

3.2.3.1 View Package

Class Name	Description
List of Inquiry Page	Displays user's submitted inquiries with assignment status and agency information. Shows which agencies are handling each inquiry with assignment dates.
View Submitted Inquiries Page	Shows all submitted inquiries for admin review with search and filter functionality for assignment management.
Assign Inquiry to Page	Provides bulk assignment interface for assigning multiple inquiries to agencies with agency selection dropdown.
Assign Inquiry Section Page	Comprehensive assignment dashboard with icon-based navigation between assignment workflows and statistics.
Add Assignment Notes & Instructions Page	Interface for adding detailed assignment notes, priority levels, and instructions when assigning inquiries to agencies.
Generate Assigned Inquiry Reports Page	Creates comprehensive reports on inquiry assignments with filtering options and export capabilities in PDF/Excel formats.
Display And Approved Page	Agency interface showing assigned inquiries with management capabilities including status updates, acceptance, and rejection functionality.

3.2.3.2 Controller Package

Class Name	Description
InquiryAssignmentController	Manages the assignment of inquiries to agencies with bulk assignment capabilities and assignment tracking.
AdminController	Handles admin operations for Module 3 including inquiry review, assignment management, and assignment report generation.
UserController	Manages public user operations for viewing assigned agencies and tracking inquiry assignment status.
AgencyController	Controls agency operations for managing assigned inquiries, processing assignments, and status updates.
InquiryController	Manages inquiry operations specific to assignment functionality including tracking and assignment status updates.

3.2.3.3 Model Package

Class Name	Description
PublicUsers	Represents public users who can view their inquiry assignment status and track which agencies are handling their submissions.
Administrator	Represents administrators who manage inquiry assignments with comprehensive assignment tracking and reporting capabilities.
AssignedInquiry	Represents the assignment relationship between inquiries and agencies with detailed assignment tracking information.
Agency	Represents agencies who receive assigned inquiries and process them with acceptance, rejection, and status update capabilities.
Inquiry	Represents inquiries with assignment functionality, tracking assignment status, assigned agencies, and processing progress.

3.2.4. Module 4: Inquiry Progress Tracking

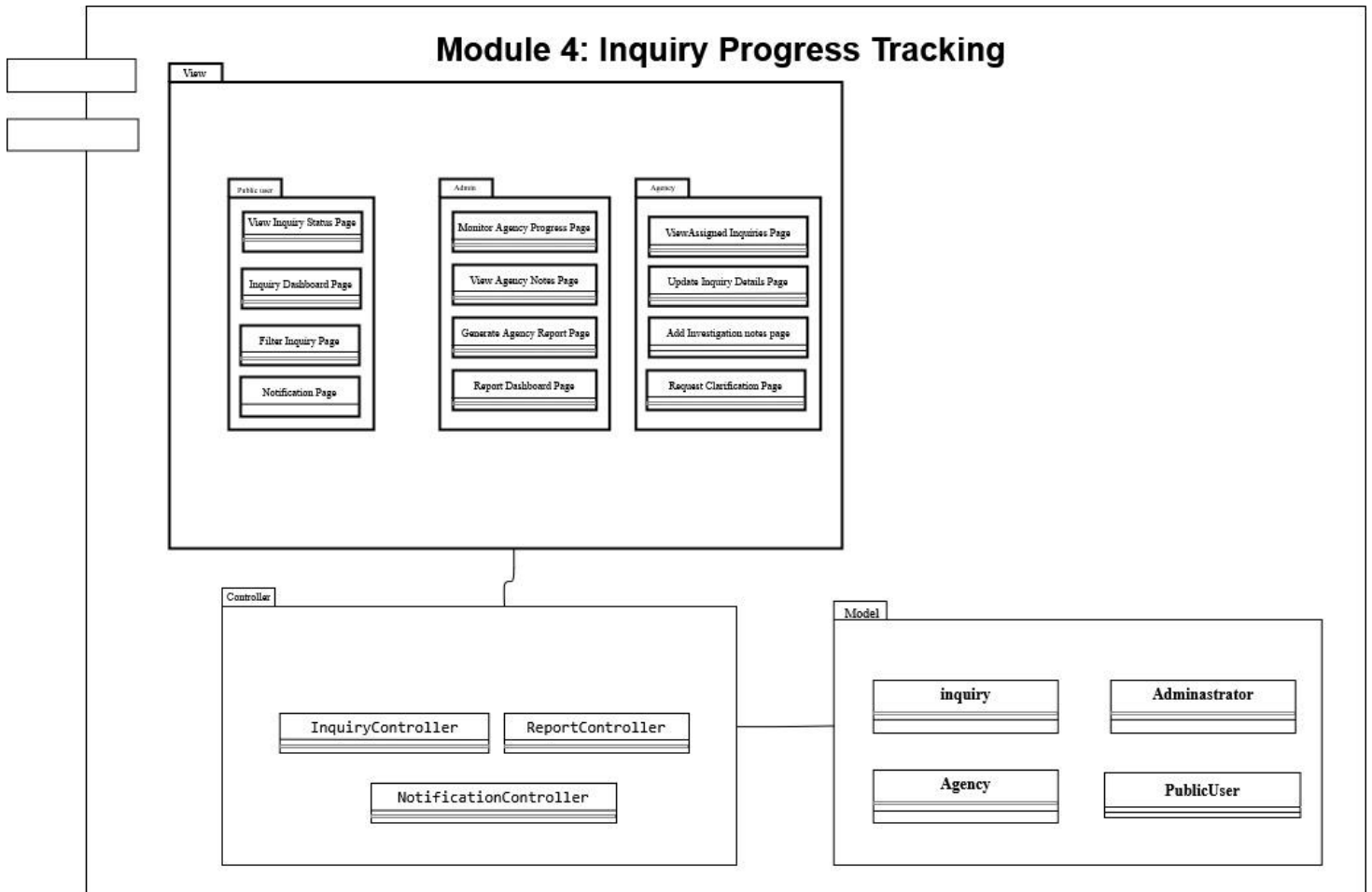


Figure 3.2.4 Component Module 4 Inquiry Progress Tracking

3.2.4.1 View Package

Class Name	Description
LandingPage	Entry point for users (public, admin, agency) into the inquiry system.
ViewInquiryStatusPage	Allows public users to check the progress of their submitted inquiries.
InquiryDashboardPage	Displays a summary of inquiries submitted by the user.
FilterInquiryPage	Enables filtering and sorting of inquiry data.
MonitorAgencyProgressPage	Allows admins to track the progress of agency-handled inquiries.
ViewNotesPage	Lets admins view notes attached to specific inquiries.

GenerateAgencyReportPage	Enables admins to compile agency-specific performance reports.
ReportDashboardPage	Overview of all system reports for admin use.
ExportReportPage	Lets admins export reports for external use or archiving.
ViewAssignedInquiriesPage	Allows agency staff to see their assigned inquiries.
UpdateInquiryStatusPage	Enables agencies to update inquiry progress and resolution.
AddDetailsPage	Lets agencies provide additional info on an inquiry.
RequestClarificationPage	Used by agencies to ask for further details from the inquirer.
InquiryCommunicationLogPage	Displays communication exchanges related to the inquiry.

3.2.4.1 Controller Package

Class Name	Description
StatusesController	Manages inquiry status updates and related logic.
ReportController	Handles report generation, dashboard data, and export functionality.
AssignedInquiryController	Controls agency interactions with assigned inquiries.

3.2.4.1 Model Package

Class Name	Description
Inquiry	Represents inquiry records with details like status, notes, and timestamps.
Administrator	Represents admin users with permissions for monitoring and reporting.
Agency	Stores data about agencies handling inquiries, such as name and contact info.
PublicUser	Represents regular users who submit inquiries and track their status.

3.3 Package Relationship

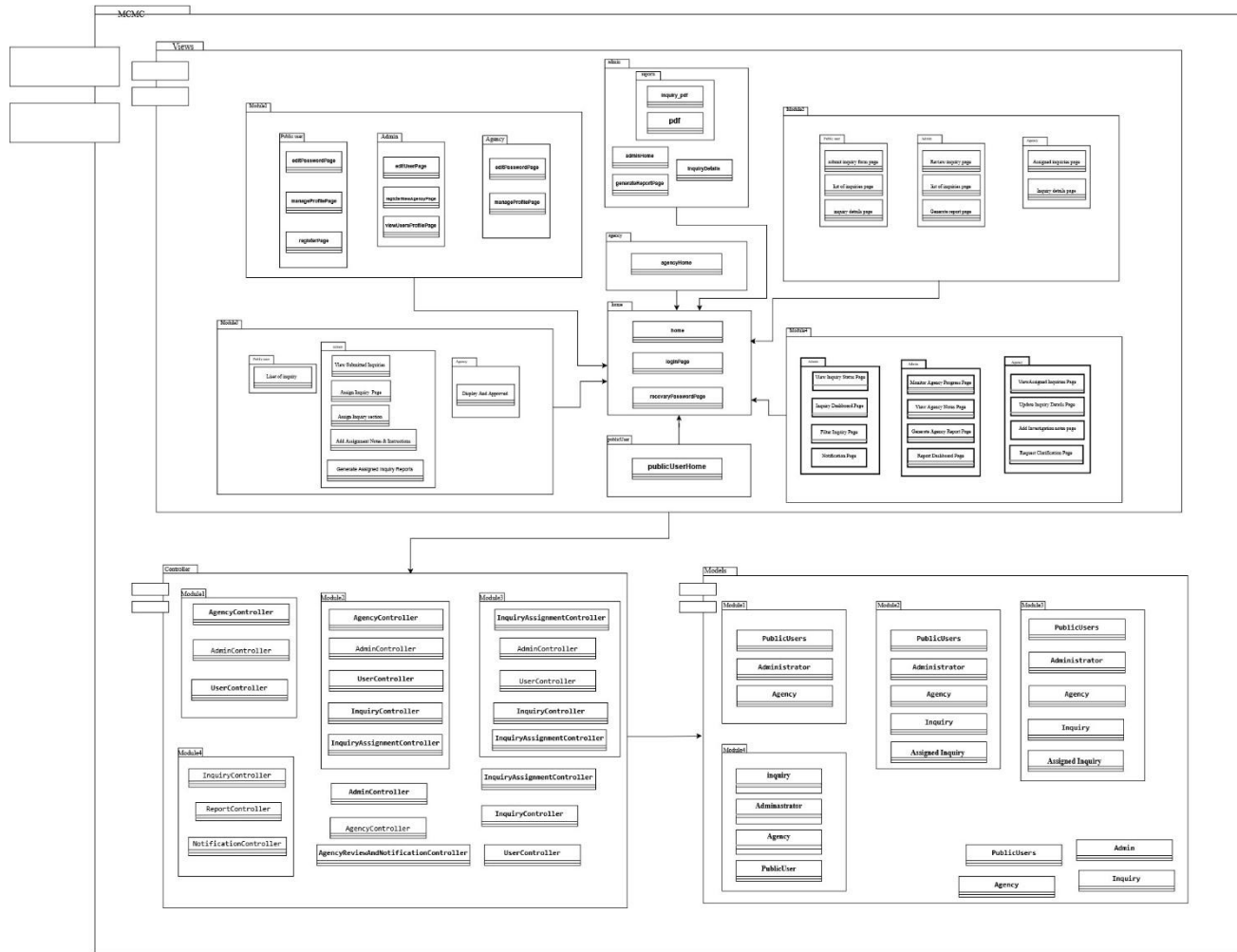
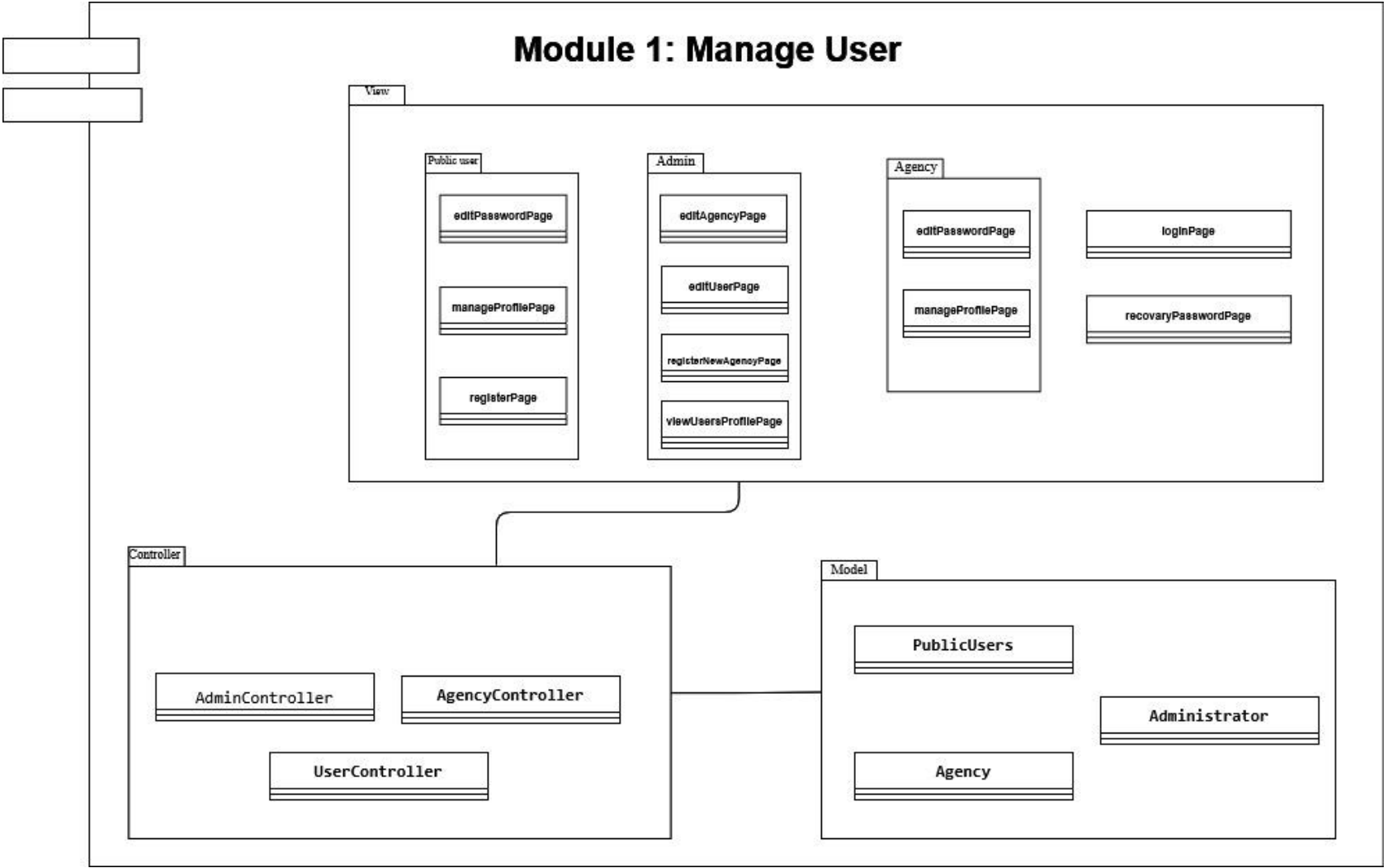


Figure 3.3 Model-View-Controller (MVC) Package Relationship for AuthenticityHub System
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4. DETAIL DESIGN

4.1 Module 1: Manage User [SDD-REQ-100]



4.1.1 PublicUsers [SDD-REQ-101]

Class Type	Entity class	
Responsibility	A model class that represents public user entities in the system and manages user-related data, authentication, and profile information.	
Attributes	Attributes Name	Attributes Type
	UserID	integer
	UserName	string
	UserEmail	string
	UserPassword	string
	UserPhoneNum	string
	UserProfilePicture	string
	created_at	timestamp
	updated_at	timestamp
Methods	Method Name	Description
	updateProfile(array \$data)	To update user profile information
	updatePassword(\$newPassword)	To update user password
	validateCurrentPassword(\$currentPassword)	To validate current password
	getAuthPassword()	To get password for authentication
	getEmailForPasswordReset()	To get email for password reset
Algorithm	1. For Profile Update (updateProfile method): - Validate updated information format and requirements - Check for duplicate email conflicts in public_users table - Begin database transaction - Update user record with new information - Commit transaction and return success status 2. For Password Update (updatePassword method): - Hash new password using bcrypt encryption - Update password field in database - Save updated record - Return success confirmation 3. For Password Validation (validateCurrentPassword method): - Retrieve current password hash from database - Compare provided password with stored hash - Return boolean validation result	

4.1.2 Agency [SDD-REQ-102]

Class Type	Entity class	
Responsibility	A model class that represents agency entities in the system and manages agency-related data and information.	
Attributes	Attributes Name	Attributes Type
	AgencyID	integer
	AgencyName	string
	AgencyUserName	string
	AgencyEmail	string
	AgencyPhoneNum	string
	AgencyPassword	string
	AgencyProfilePicture	string
	first_login	boolean
	password_changed_at	timestamp
Methods	Method Name	Description
	updateProfile(array \$data)	To update agency profile information
	updatePassword(\$newPassword)	To update agency password
	validateCurrentPassword(\$currentPassword)	To validate current password
	checkPassword(\$password)	To check password for login
Algorithm	1. For Profile Update (updateProfile method): - Validate updated information format and requirements - Check for duplicate username/email conflicts in agencies table - Begin database transaction - Update agency record with new information - Commit transaction and return success status 2. For Password Update (updatePassword method): - Hash new password using bcrypt encryption - Update password field in database - Update password_changed_at timestamp - Save updated record 3. For Password Check (checkPassword method): - Retrieve current password hash from database - Compare provided password with stored hash using Hash::check - Return boolean result	

4.1.3 Administrator [SDD-REQ-103]

Class Type	Entity class	
Responsibility	A model class that represents administrator entities with elevated privileges for system management operations.	
Attributes	Attributes Name	Attributes Type
	AdminID	integer
	AdminName	string
	AdminEmail	string
	AdminPassw	string
	AdminRole	string
	AdminPhone	string
	AdminAddre	string
	AdminUserN	string
	created_at	timestamp
	updated_at	timestamp
Methods	Method Name	Description
	checkPassword(\$password)	To check admin password for authentication
	updateProfile(array \$data)	To update admin profile information
	updatePassword(\$new Password)	To update admin password
	1. For Password Check (checkPassword method): a. Retrieve current password hash from database b. Compare provided password with stored hash using Hash::check c. Return boolean authentication result 2. For Profile Update (updateProfile method): a. Validate updated information format and requirements b. Begin database transaction c. Update admin record with new information d. Commit transaction and return success status	

4.1.4 UserController [SDD-REQ-104]

Class Type	Controller class	
Responsibility	A controller that handles user-related operations including registration, profile management, and authentication for public users.	
	Attributes Name	Attributes Type
	user->UserName	string
	user->UserEmail	string
	user->UserPassword	string
	user->UserID	integer
Methods	Method Name	Description
	showRegistrationForm()	To display user registration form
	register(Request \$request)	To handle user registration process
	showLoginForm()	To display login form
	login(Request \$request)	To handle user authentication
	updateProfile(Request \$request)	To update user profile information
	verifyPassword(Request \$request)	To verify current password
	updatePassword(Request \$request)	To update user password
	logout(Request \$request)	To handle user logout
Algorithm	1. For User Registration (register method): - Validate registration form data format and requirements - Check for existing email in database - Hash password using Laravel's bcrypt encryption - Create user record with validated data - Redirect to login with success message 2. For User Login (login method): - Validate login credentials format - Attempt admin login first through AdminController - Try agency login with proper password checking - Finally attempt public user login - Authenticate and redirect based on user type 3. For Profile Update (updateProfile method): - Validate updated information format and requirements - Check for duplicate email conflicts excluding current user - Handle profile picture upload if provided - Update user record with new validated information - Return success message	

4.1.5 AdminController [SDD-REQ-105]

Class Type	Controller class	
Responsibility	A controller that handles administrative operations including user management, agency management, and system oversight functions.	
Attributes	Attributes Name	Attributes Type
	admin->AdminName	string
	admin->AdminEmail	string
	admin->AdminRole	string
	admin->AdminID	integer
Methods	Method Name	Description
	showHome()	To display admin dashboard
	showUsers(Request \$request)	To display user management page
	editUser(\$id)	To show user edit form
	updateUser(Request \$request, \$id)	To update user information
	editAgency(\$id)	To show agency edit form
	updateAgency(Request \$request, \$id)	To update agency information
	showAgencyRegistrationForm()	To show agency registration form
	storeAgency(Request \$request)	To register new agency
	deleteUser(\$id)	To delete user account
	deleteAgency(\$id)	To delete agency account
	showReports()	To show reports page
	generateReports(Request \$request)	To generate system reports

Algorithm	1. For User Management (showUsers method): a. Retrieve user list with pagination parameters b. Apply search filters for name, email c. Return paginated user results to view 2. For Agency Registration (storeAgency method): a. Validate agency data format and requirements b. Check for duplicate agency codes/usernames c. Create agency record with validated data d. Generate temporary password e. Send credentials via email 3. For User Update (updateUser method): a. Validate updated user information format b. Check for duplicate email conflicts c. Update user record with new information d. Return success status
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4.1.6 AgencyController [SDD-REQ-106]

Class Type	Entity class	
Responsibility	A model class that represents agency entities in the system and manages agency-related data and information.	
Attributes	Attributes Name	Attributes Type
	agency->AgencyName	string
	agency->AgencyUserName	string
	agency->AgencyEmail	string
	agency->AgencyID	integer
Methods	Method Name	Description
	showProfile()	To display agency profile page
	updateProfile(Request \$request)	To update agency profile information
	showSecurity()	To display password management page
	verifyPassword(Request \$request)	To verify current password
	updatePassword(Request \$request)	To update agency password
	showHome()	To display agency dashboard
Algorithm	1. For Profile Update (updateProfile method): - Validate updated agency information format - Check for username/email conflicts with other agencies - Handle profile picture upload if provided - Update agency record with validated information - Return success response 2. For Password Update (updatePassword method): - Validate new password meets requirements - Hash new password using bcrypt - Update password in database through model - Clear password verification session - Return success confirmation	

4.1.7 RegisterPage [SDD-REQ-107]

Class Type	Boundary class	
Responsibility	A user registration interface that provides functionality for new public users to create accounts by providing their personal information and credentials.	
Attributes	Attributes Name	Attributes Type
	UserName	string
	UserEmail	string
	UserPassword	string
	UserPassword_confirmation	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Registration Form Display: a. Render HTML form with UserName, UserEmail, UserPassword, UserPassword_confirmation fields b. Include @csrf token for security c. Apply input-smooth CSS classes for animations d. Use @error directives for validation error display e. Include JavaScript for form animations and help modal	

4.1.8 ManageProfilePage (PublicUser) [SDD-REQ-108]

Class Type	Boundary class	
Responsibility	A profile management interface that enables logged-in public users to view, update, and manage their personal profile information including name, email, matric number, and profile	
Attributes	Attributes Name	Attributes Type
	UserName	string
	UserEmail	string
	UserPhoneNum	string
	UserProfilePicture	file
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Profile Display: a. Authenticate user session via Auth::user() b. Populate form fields with Auth::user() data c. Display current profile picture or default avatar d. Include file upload with preview functionality via JavaScript e. Handle form submission with loading animations	

4.1.9 EditPasswordPage (PublicUser) [SDD-REQ-109]

Class Type	Boundary class	
Responsibility	A password change interface that allows authenticated public users to securely update their account password by providing current and new password credentials.	
Attributes	Attributes Name	Attributes Type
	current_password	string
	new_password	string
	new_password_confirmation	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Password Change Process: a. Check session('password_verified') to determine display state b. Show verification form if not verified c. Show update form if verified d. Include form animations and session management	

4.1.10 RegisterNewAgencyPage [SDD-REQ-110]

Class Type	Boundary class	
Responsibility	An admin-only interface that allows administrators to register and add new agencies to the system with relevant agency details and information.	
Attributes	Attributes Name	Attributes Type
	AgencyName	string
	AgencyEmail	string
	AgencyUserName	string
	AgencyPassword	string
	AgencyPassword_confirmation	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Agency Registration Form: a. Verify admin authentication via middleware b. Render form with agency registration fields c. Include @csrf token for security d. Handle form validation errors with @error directives	

4.1.11 ViewUsersProfilePage [SDD-REQ-111]

Class Type	Boundary class	
Responsibility	An admin dashboard interface that displays profile information of all registered users and agencies, providing administrators with a comprehensive overview of system users.	
Attributes	Attributes Name	Attributes Type
	users	collection
	agencies	collection
	search	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Users List Display: a. Display paginated users and agencies data in tables b. Include search form for filtering results c. Provide edit and delete links for each record d. Show pagination controls	

4.1.12 editAgencyPage [SDD-REQ-112]

Class Type	Boundary class	
Responsibility	An admin interface that allows administrators to modify and update existing agency information and details in the system.	
Attributes	Attributes Name	Attributes Type
	AgencyName	string
	AgencyEmail	string
	AgencyUserName	string
	AgencyPhoneNum	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Agency Edit Form: a. Retrieve agency data and populate form fields b. Include @csrf token and @method('PUT') c. Handle validation errors and success messages d. Provide save and cancel options	

4.1.13 editUserPage [SDD-REQ-113]

Class Type	Boundary class	
Responsibility	An admin interface that enables administrators to edit and update public user account information and profile details.	
Attributes	Attributes Name	Attributes Type
	UserName	string
	UserEmail	string
	UserPhoneNum	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For User Edit Form: a. Retrieve user data and populate form fields b. Include @csrf token and @method('PUT') c. Handle validation errors and success messages d. Provide save and cancel options	

4.1.14 LoginPage [SDD-REQ-114]

Class Type	Boundary class	
Responsibility	A login interface that provides authentication functionality for all user types (public users, MCMC staff, and agency staff) to access the system using their credentials.	
Attributes	Attributes Name	Attributes Type
	login	string
	password	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Login Form Display: a. Render login form with email/username and password fields b. Include forgot password link to route('password.recovery') c. Include @csrf token for security d. Handle authentication errors display	

4.1.15 RecoveryPasswordPage [SDD-REQ-115]

Class Type	Boundary class	
Responsibility	A password recovery interface that allows users to reset their forgotten passwords by providing their email address or username for password reset instructions.	
Attributes	Attributes Name	Attributes Type
	email	string
	phone	string
	password	string
	password_confirmation	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Recovery Form Display: a. Display different forms based on recovery type b. Include agency phone and password setup form c. Include email recovery form d. Handle form validation and submissions	

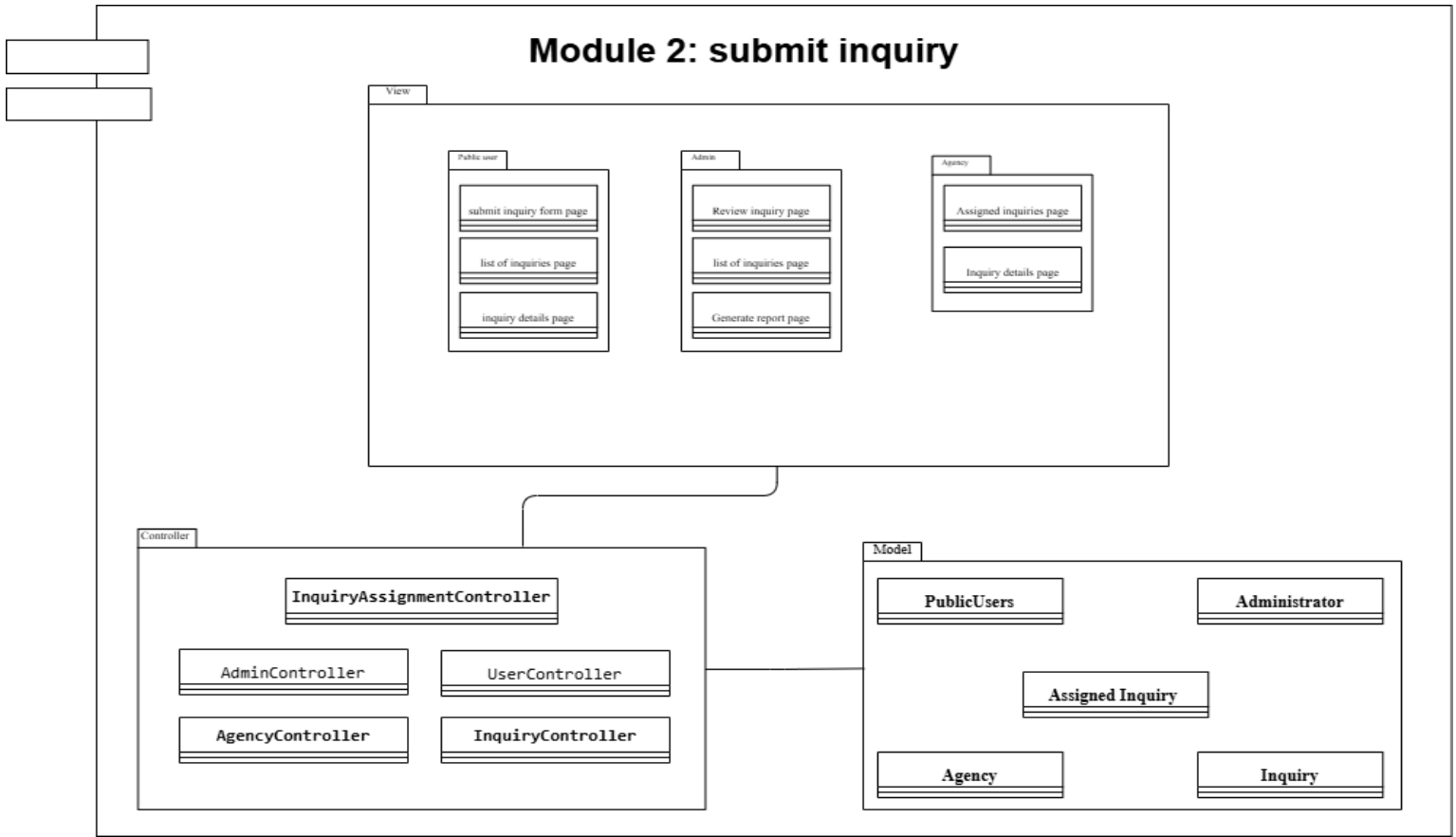
4.1.16 ManageProfilePage (Agency) [SDD-REQ-116]

Class Type	Boundary class	
Responsibility	A profile management interface that enables logged-in agency staff to view, update, and manage their agency profile information including agency name, contact details, and profile picture.	
Attributes	Attributes Name	Attributes Type
	AgencyName	string
	AgencyEmail	string
	AgencyPhoneNum	string
	AgencyUserName	string
	AgencyProfilePicture	file
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Agency Profile Display: - Authenticate agency session via Auth::user() - Populate form fields with current agency data - Include file upload capability for profile picture - Handle form submission with validation	

4.1.17 EditPasswordPage (Agency) [SDD-REQ-117]

Class Type	Boundary class	
Responsibility	A password change interface that allows authenticated agency staff to securely update their account password, with special handling for first-time login password changes from temporary	
Attributes	Attributes Name	Attributes Type
	current_password	string
	new_password	string
	new_password_confirm	string
Methods	Method Name	Description
	None (Blade Template)	This is a view file, not a class with methods
Algorithm	1. For Agency Profile Display: - Authenticate agency session via Auth::user() - Populate form fields with current agency data - Include file upload capability for profile picture - Handle form submission with validation	

4.2 Module 2: Manage Inquiry Form Submission [SDD-REQ-200]



4.2.1 submitInquiryFormView [SDD-REQ-201]

Class Type	Boundary class	
Responsibility	A view interface that displays the main inquiry submission form with input fields for inquiry details and file attachments.	
Attributes	Attributes Name	Attributes Type
	InquiryTitle	string
	InquiryDescription	text
	InquiryNewsSource	string
	InquiryEvidence	file
	validationErrors	array
	old_values	array

Methods	Method Name	Description
	route('submit.inquiry')	To handle form submission for inquiry creation
	old('InquiryTitle')	To retrieve old input value for inquiry title
	old('InquiryDescription')	To retrieve old input value for description
	old('InquiryNewsSource')	To retrieve old input value for news source
	@error('field')	To display validation errors for specific fields
Algorithm	1. Display inquiry submission form: a. Load form with fields: InquiryTitle, InquiryDescription, InquiryNewsSource, b. Display file upload for InquiryEvidence, c. Show validation errors if any exist, d. Retain old input values using old() helper 2. Handle form interaction: a. Validate required fields (title, description, source), b. Check file upload constraints for evidence, c. Display real-time validation feedback 3. On form submission: a. Submit to route('submit.inquiry'), b. Show loading state with spinner animation, c. Handle success/error responses	

4.2.2 listOfInquiriesView [SDD-REQ-202]

Class Type	Boundary class	
Responsibility	A view interface that displays a paginated list of user's submitted inquiries with status tracking.	
Attributes	Attributes Name	Attributes Type
	inquiries	collection
	InquiryID	integer
	InquiryTitle	string
	InquiryStatus	string
	created_at	datetime
	pagination	object
Methods	Method Name	Description
	route('inquiries.index')	To display inquiry list page
	route('inquiry.details', id)	To view specific inquiry details
	formatDate()	To format submission dates

Algorithm	1. Load user inquiries: a. Fetch inquiries belonging to authenticated user, b. Display InquiryID, InquiryTitle, InquiryStatus, created_at, c. Apply pagination if needed 2. Display inquiry cards: a. Show inquiry status with color-coded badges, b. Display submission date in readable format, c. Provide "View Details" links 3. Handle empty state: a. Show message if no inquiries found, b. Provide link to submit new inquiry
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4.2.3 inquiryDetailsView [SDD-REQ-203]

Class Type	A view interface that displays detailed information of a specific inquiry including all related data.	
Responsibility	A view interface that displays a paginated list of user's submitted inquiries with status tracking.	
Attributes	Attributes Name	Attributes Type
	InquiryID	integer
	InquiryTitle	string
	InquiryDescription	text
	InquiryNewsSource	string
	InquiryEvidence	string
	InquiryStatus	string
	created_at	datetime
	AgencyName	string
Methods	Method Name	Description
	asset('storage/' . evidence)	To generate download links for evidence files
	route('inquiries.index')	To navigate back to inquiry list

Algorithm	1. Load inquiry details: a. Fetch inquiry by InquiryID, b. Verify user ownership of inquiry, c. Load related agency information if assigned 2. Display comprehensive information: a. Show all inquiry fields with proper formatting, b. Display evidence file with download link, c. Show current status and assignment details 3. Handle file downloads: a. Generate secure links to evidence files, b. Open files in new tab for viewing
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4.2.4 InquiryAssignmentController [SDD-REQ-204]

Class Type	Control class	
Responsibility	A controller class that manages the assignment of inquiries to appropriate agencies based on admin decisions.	
Attributes	Attributes Name	Attributes Type
	AgencyID	integer
	InquiryID	integer
	assignment_date	datetime
	AdminID	integer
Methods	Method Name	Description
	assignInquiry()	To assign inquiry to selected agency
	updateInquiryStatus()	To update inquiry status after assignment
Algorithm	1. Process assignment request: a. Validate inquiry and agency IDs, b. Check if inquiry is pending assignment, c. Verify agency availability 2. Execute assignment: a. Update inquiry record with AgencyID, b. Set assignment_date to current timestamp, c. Change InquiryStatus to "Under Investigation" 3. Create assignment record: a. Log assignment in database, b. Send notification to agency	

4.2.5 AdminController [SDD-REQ-205]

Class Type	Control class
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Responsibility	A controller class that handles administrative functions for inquiry management and system oversight.	
Attributes	Attributes Name	Attributes Type
	AdminID	integer
	AdminName	string
	inquiry_details	object
	available_agencies	collection
Methods	Method Name	Description
	route('admin.inquiry.details', id)	To view detailed inquiry information
	route('admin.assignment.interface')	To access assignment interface
	route('admin.inquiries')	To view all inquiries
Algorithm	1. Administrative overview: a. Display pending inquiries requiring assignment, b. Show inquiry statistics and metrics, c. Provide quick assignment actions 2. Inquiry review process: a. Load inquiry with all details, b. Display evidence files and descriptions, c. Provide assignment and status update options 3. Assignment management: a. List available agencies for assignment, b. Process assignment requests, c. Update inquiry status and tracking	

4.2.6 Inquiry [SDD-REQ-206]

Class Type	Entity class	
Responsibility	An entity class representing inquiry records with complete information including content, status, and tracking data.	
Attributes	Attributes Name	Attributes Type
	user()	To get related user who submitted inquiry
	agency()	To get assigned agency
	admin()	To get assigned administrator
	user()	To get related user who submitted inquiry
	agency()	To get assigned agency
	admin()	To get assigned administrator
	user()	To get related user who submitted inquiry
	agency()	To get assigned agency

	admin()	To get assigned administrator
	user()	To get related user who submitted inquiry
	agency()	To get assigned agency
	admin()	To get assigned administrator
	user()	To get related user who submitted inquiry
Methods	Method Name	Description
	user()	To get related user who submitted inquiry
	agency()	To get assigned agency
	admin()	To get assigned administrator
Algorithm	1. Inquiry creation: a. Generate unique InquiryID, b. Set InquiryStatus to "Pending", c. Store InquiryTitle, InquiryDescription, InquiryNewsSource, d. Handle InquiryEvidence file upload 2. Status management: a. Track status changes: Pending → Under Investigation → Verified/Rejected, b. Update timestamps for each status change, c. Maintain audit trail 3. Assignment tracking: a. Record AgencyID when assigned, b. Set assignment date timestamp, c. Link to AdminID who made assignment	

4.2.7 PublicUser [SDD-REQ-207]

Class Type	Entity class	
Responsibility	An entity class representing public users who submit inquiries with contact information.	
Attributes	Attributes Name	Attributes Type
	UserID	integer
	UserName	string
	UserEmail	string
	UserPhoneNum	string
	password	string
	created_at	datetime
Methods	Method Name	Description

	inquiries()	To get all inquiries submitted by user
	hasVerifiedEmail()	To check email verification status
Algorithm	1. User registration: a. Validate userEmail uniqueness, b. Hash password securely, c. Generate UserID 2. Inquiry submission: a. Authenticate user session, b. Associate inquiry with UserID, c. Track submission history	

4.2.8 Agency [SDD-REQ-208]

Class Type	Entity class	
Responsibility	An entity class representing government agencies that handle specific inquiry types.	
Attributes	Attributes Name	Attributes Type
	AgencyID	integer
	AgencyName	string
	AgencyUserName	string
	AgencyEmail	string
	AgencyPhoneNum	string
	password	string
	created_at	datetime
Methods	Method Name	Description
	assignedInquiries()	To get inquiries assigned to agency
	route('agency.view.display.inquiry')	To access agency inquiry management
Algorithm	1. Agency setup: a. Register with unique AgencyUserName, b. Set AgencyName and contact details, c. Configure investigation capabilities 2. Inquiry processing: a. Receive assigned inquiries, b. Update InquiryStatus during investigation, c. Provide verification results	

4.2.9 Administrator [SDD-REQ-209]

Class Type	Entity class	
Responsibility	An entity class representing system administrators with full privileges for inquiry management.	
Attributes	Attributes Name	Attributes Type
	AdminID	integer
	AdminName	string
	AdminEmail	string
	password	string
	created_at	datetime
Methods	Method Name	Description
	assignedInquiries()	To get inquiries managed by admin
	route('admin.home')	To access admin dashboard
Algorithm	1. Administrative oversight: a. Monitor all inquiry submissions, b. Assign inquiries to appropriate agencies, c. Track investigation progress 2. System management: a. Manage agency registrations, b. Generate system reports, c. Handle escalated cases	

4.2.10 AssignedInquiry [SDD-REQ-210]

Class Type	Entity class	
Responsibility	An entity class representing the assignment relationship between inquiries, agencies, and administrators.	
Attributes	Attributes Name	Attributes Type
	assignmentId	integer
	InquiryID	integer
	AgencyID	integer
	AdminID	integer
	assignment_date	datetime
	status	string

Methods	Method Name	Description
	inquiry()	To get related inquiry
	agency()	To get assigned agency
	admin()	To get assigning administrator
Algorithm	1. Create assignment: a. Link InquiryID with AgencyID, b. Record AdminID who made assignment, c. Set assignment_date timestamp 2. Track assignment status: a. Monitor investigation progress, b. Update status as inquiry progresses, c. Maintain assignment history	

4.2.11 InquiryController [SDD-REQ-211]

Class Type	Control class	
Responsibility	A controller class that handles core inquiry operations including creation, modification, status updates, and inquiry lifecycle management.	
Attributes	Attributes Name	Attributes Type
	inquiryValidationRules	array
	statusWorkflow	object
	inquiryTypes	array
Methods	Method Name	Description
	createInquiry()	To create new inquiry records
	updateInquiryStatus()	To update inquiry processing status
	validateInquiryData()	To validate inquiry information

Algorithm	1. Create new inquiry: a. Validate submitted inquiry data, b. Generate unique inquiry ID, c. Set initial status and timestamps, d. Store in database 2. Status management: a. Validate status transition rules, b. Update inquiry status, c. Log status change history, d. Trigger notifications 3. Data validation: a. Check required fields, b. Validate data formats, c. Ensure business rule compliance 4. Lifecycle management: a. Track inquiry from creation to closure, b. Manage automated workflows, c. Handle inquiry archival
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4.2.12 reviewInquiryPageView [SDD-REQ-212]

Class Type	Boundary class	
Responsibility	An admin view interface that displays individual inquiry details for review and status management.	
Attributes	Attributes Name	Attributes Type
	inquiry	object
	InquiryID	integer
	InquiryTitle	string
	InquiryDescription	text
	InquiryNewsSource	string
	InquiryEvidence	string
	InquiryStatus	string
	created_at	datetime
	user	object
Methods	Method Name	Description
	route('admin.inquiry.details/{id}')	To display inquiry review details
	asset('storage/' . evidence)	To display evidence files
	route('admin.inquiries')	To return to inquiry list

Algorithm	1. Load inquiry for review: a. Fetch complete inquiry details by InquiryID, b. Load related user information, c. Display evidence files with download capability, d. Show current status and submission date 2. Display review interface: a. Show all inquiry fields in detail, b. Display evidence files and attachments, c. Show user contact information 3. Admin review capabilities: a. View complete inquiry information, b. Review evidence and documentation, c. Navigate back to inquiry list
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4.2.13 listOfInquiriesPageView [SDD-REQ-213]

Class Type	Boundary class	
Responsibility	An admin view interface that displays a searchable list of all submitted inquiries with review functionality.	
Attributes	Attributes Name	Attributes Type
	inquiries	collection
	InquiryID	integer
	InquiryTitle	string
	InquiryNewsSource	string
	InquiryStatus	string
	created_at	datetime
	search	string
	status	string
Methods	Method Name	Description
	route('admin.inquiries')	To display inquiry list
	route('admin.inquiry.details', id)	To review specific inquiry
	request('search')	To handle search functionality
	request('status')	To filter by status

Algorithm	1. Display inquiry list: a. Load all submitted inquiries, b. Apply search filters on title, source, or ID, c. Filter by inquiry status if selected, d. Sort by submission date 2. Search and filter functionality: a. Search across inquiry fields, b. Filter by status options, c. Clear filters option 3. Review navigation: a. Display "Review" button for each inquiry, b. Show status badges, c. Navigate to detailed review page
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4.2.14 generateReportPageView [SDD-REQ-214]

Class Type	Boundary class	
Responsibility	An admin view interface for generating reports on submitted inquiries with various filtering and export options.	
Attributes	Attributes Name	Attributes Type
	report_type	string
	date_from	date
	date_to	date
	status	string
	format	string
	totalInquiries	integer
	statusCounts	array
	inquiries	collection
Methods	Method Name	Description
	route('admin.reports')	To display reports page
	route('admin.generate.inquiry.reports')	To generate reports
	generateInquiryReports()	To process report creation

Algorithm	1. Report configuration: a. Set date range for inquiry filtering, b. Select report type (Summary, Detailed, Statistics), c. Choose export format (PDF, Excel), d. Apply status filters 2. Data collection and analysis: a. Filter inquiries by date range, b. Count inquiries by status, c. Calculate submission statistics, d. Generate trends and patterns 3. Report generation: a. Compile inquiry data based on filters, b. Generate charts and statistics, c. Export in selected format, d. Provide downloadable report file
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4.2.15 assignedInquiriesPageView [SDD-REQ-215]

Class Type	Boundary class	
Responsibility	An agency view interface that displays inquiries assigned to the agency with management capabilities including status updates and rejection functionality.	
Attributes	Attributes Name	Attributes Type
	agency	object
	inquiries	collection
	statusCounts	array
	InquiryID	integer
	InquiryTitle	string
	InquiryDescription	text
	InquiryNewsSource	string
	InquiryEvidence	string
	InquiryStatus	string
	InquiryPriority	string
	assignment_date	datetime
	user	object
Methods	Method Name	Description
	route('agency.view.display.inquiry')	To display assigned inquiries page
	showInquiryDetails(inquiryId)	To show detailed inquiry modal
	rejectInquiry(inquiryId)	To reject an inquiry
	filterInquiries()	To filter inquiries by status/priority

	clearFilters()	To clear all applied filters
Algorithm	- Load assigned inquiries: - Fetch inquiries where AgencyID matches current agency - Order by priority: Urgent, High, Normal, then by assignment_date desc - Load with user, agency, and admin relationships - Calculate status counts for dashboard - Display inquiry management interface: - Show summary cards with total, under_investigation, verified_true, identified_fake counts - Display filter controls for status and priority - Show inquiry cards with InquiryID, InquiryTitle, status badges, priority levels - Include action buttons: "Show Full Details", "Reject Inquiry" - Handle inquiry interactions: - Show detailed modal with complete inquiry information - Display evidence files with download links using asset('storage/') - Process rejection with AJAX to update InquiryStatus - Apply real-time filtering by status and priority - Status and priority management: - Display status badges: Pending, Under Investigation, Verified as True, Identified as Fake - Show priority badges: Urgent, High, Normal - Filter by status: All, Pending, Under Investigation, etc. - Filter by priority: All Priorities, Urgent, High, Normal	

4.2.16 inquiryDetailsPageView [SDD-REQ-216]

Class Type	Boundary class	
Responsibility	An agency view interface that displays complete details of a specific inquiry assigned to the agency with full information and action capabilities.	
Attributes	Attributes Name	Attributes Type
	inquiry	object
	agency	object
	InquiryID	integer
	InquiryTitle	string
	InquiryDescription	text
	InquiryNewsSource	string
	InquiryEvidence	string
	InquiryStatus	string
	InquiryPriority	string
	assignment_date	datetime
	created_at	datetime
	user	object

	evidenceFiles	array
Methods	Method Name	Description
	route('agency.inquiry.details', id)	To display specific inquiry details
	asset('storage/' . evidence)	To generate evidence file URLs
	route('agency.view.display_inquiry')	To return to inquiry list
	formatDate()	To format dates for display
Algorithm	1. Load complete inquiry details: - Fetch inquiry by InquiryID where AgencyID matches current agency - Load with user, agency, and admin relationships - Verify inquiry belongs to current agency - Handle not found or unauthorized access 2. Display comprehensive information: - Show inquiry details: ID, Title, Description, NewsSource, Status, Priority - Display assignment information: assignment_date, agency details - Show submitter information: UserName, UserEmail, UserPhoneNum - Format dates using created_at and assignment_date 3. Handle evidence files: - Parse InquiryEvidence (JSON string or single file) - Display evidence items with file icons - Generate download links using asset('storage/' . file) - Open files in new tab for viewing 4. AJAX compatibility: - Handle AJAX requests for modal display - Return JSON response with inquiry and agency data - Support both full page and modal view modes - Provide seamless user experience	

4.2.17 UserController [SDD-REQ-217]

Class Type	Control class	
Responsibility	A controller class that manages public user operations including inquiry submission, viewing, authentication, and profile management.	
Attributes	Attributes Name	Attributes Type
	userSession	object
	loginInput	string
	password	string
	validationRules	array
	phone_verification_age ncv_id	integer

	agency_id	integer
	previous_url	string
Methods	Method Name	Description
	showRegistrationForm()	To display user registration form
	showLoginForm()	To display login form
	register()	To handle user registration
	login()	To handle authentication for users, agencies, and admins
	attemptAdminLogin()	To attempt admin authentication
	updateAgencyPhoneAndPassword()	To update agency phone and password during setup
	verifyPassword()	To verify current password for changes
	logout()	To handle user logou
Algorithm	- User registration process: - Validate UserName, UserEmail, UserPassword with confirmation - Check email uniqueness in public_users table - Hash password using Hash::make() - Create new PublicUser record - Redirect to login with success message - Multi-user authentication (login): - Validate login input (email or username) and password - Try admin login first using attemptAdminLogin() - Try agency login with email/username detection - Check agency password using checkPassword() method - Handle agency phone verification if needed - Try public user login as fallback - Store appropriate session data and redirect - Agency phone and password setup: - Validate phone and password with confirmation - Apply password regex for complexity - Get agency from phone_verification_agency_id session - Hash password and update AgencyPhoneNum, AgencyPassword - Clear verification session and log agency in - Session and security management: - Store previous_url for back button functionality - Mark login_successful in session - Handle session invalidation on logout - Clear sensitive session data appropriately	

4.2.18 AgencyController [SDD-REQ-218]

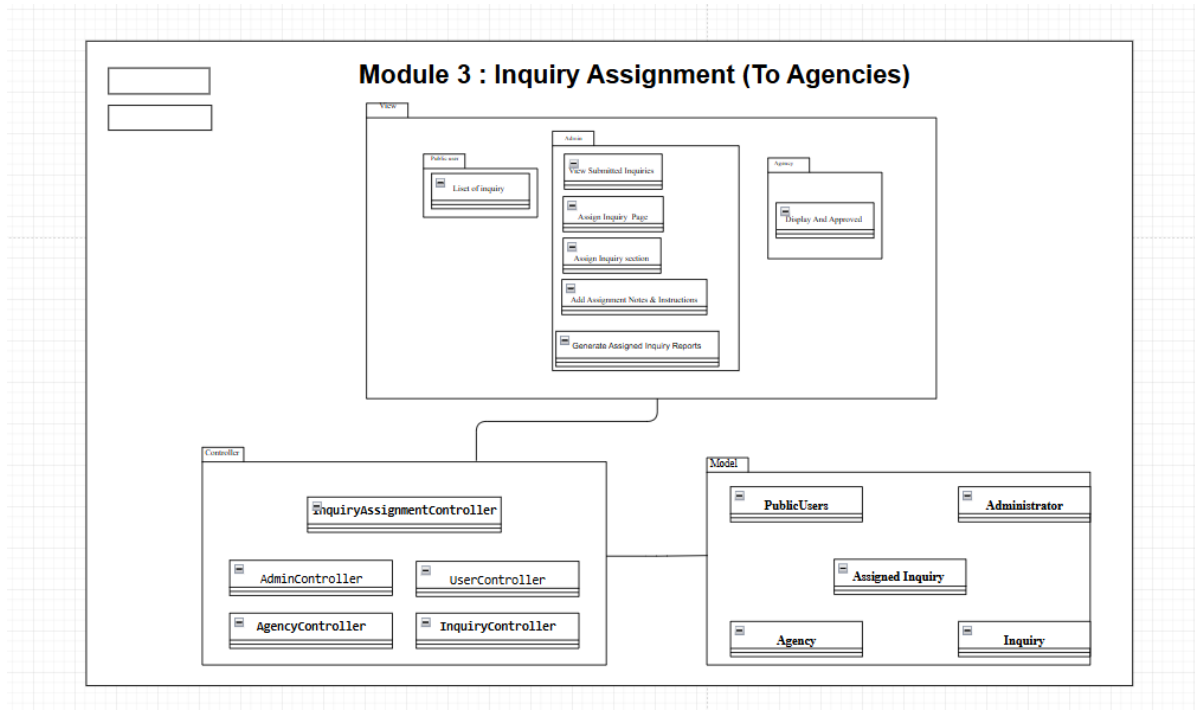
Class Type	Control class	
Responsibility	A controller class that manages agency operations including dashboard, inquiry management, authentication, and status updates.	
Attributes	Attributes Name	Attributes Type

	agency_id	integer
	agency_name	string
	statusCounts	array
	inquiries	collection
	AgencyID	integer
	InquiryStatus	string
	phone_verification_agency_id	integer
Methods	Method Name	Description
	showHome()	To display agency dashboard with statistics
	attemptLogin()	To handle agency authentication
	updateAgencyPhoneAndPassword()	To update phone and password during setup
	showViewAndDisplayInquiry()	To display assigned inquiries
	showInquiryDetails()	To show specific inquiry details (AJAX compatible)
	rejectInquiry()	To reject an inquiry with reason
	acceptInquiry()	To accept an inquiry with comments

Algorithm

1. Agency dashboard display:
 - a. Verify agency_id exists in session
 - b. Fetch agency record from database
 - c. Calculate statusCounts: total, under_investigation, verified_true, identified_fake
 - d. Return agency home view with statistics
2. Agency authentication:
 - a. Find agency by login input (email or username)
 - b. Verify password using checkPassword() method
 - c. Check if AgencyPhoneNum is empty
 - d. Handle phone verification redirect if needed
 - e. Store agency_id and agency_name in session
3. Inquiry management:
 - a. Load inquiries where AgencyID matches current agency
 - b. Order by priority: Urgent, High, Normal, then by assignment_date
 - c. Include user, agency, admin relationships
 - d. Calculate status counts for dashboard
4. Inquiry details handling:
 - a. Verify inquiry belongs to current agency
 - b. Load complete inquiry with relationships
 - c. Handle AJAX requests with JSON response
 - d. Return view for regular requests
5. Inquiry status updates:
 - a. Validate inquiry ownership by agency
 - b. For rejection: validate reason and comments
 - c. Update InquiryStatus and StatusComments
 - d. Set ProcessedAt timestamp
 - e. Return JSON response for AJAX calls
6. Phone and password setup:
 - a. Validate phone format and password complexity
 - b. Hash password using Hash::make()
 - c. Update AgencyPhoneNum and AgencyPassword
 - d. Clear phone_verification_agency_id session
 - e. Log agency in with success message

4.2 Module 3: Inquiry Assignment [SDD-REQ-300]



4.3.1 listOfInquiryView [SDD-REQ-301]

Class Type	Boundary class	
Responsibility	A public user view interface that displays a list of the user's submitted inquiries with their current assignment status and agency information.	
Attributes	Attributes Name	Attributes Type
	userInquiries	collection
	InquiryID	integer
	InquiryTitle	string
	InquiryStatus	string
	created_at	datetime
	AgencyID	integer
	AgencyName	string
	assignment_date	datetime
	user	object
	agency	object

Methods	Method Name	Description
	route('inquiries.index')	To display user's inquiry list
	route('inquiry.details', id)	To view specific inquiry details
	asset('storage/' . evidence)	To display evidence files
	Auth::user()	To get authenticated user information
	formatDate()	To format submission and assignment dates
Algorithm	- Load user's submitted inquiries: - Fetch inquiries where UserID matches authenticated user - Load with agency relationship to show assignment details - Order by created_at desc to show newest first - Include inquiry status and assignment information - Display inquiry list interface: - Show table with columns: Inquiry ID, Title, Submission Date, Status, Assigned Agency, Action - Display status badges with color coding: Pending (yellow), Under Investigation (blue), etc. - Show assigned agency name or "Not Assigned" if AgencyID is null - Provide "View Details" action button for each inquiry - Handle assignment status display: - Check if inquiry has been assigned to agency (AgencyID not null) - Display agency name if assigned, otherwise show "Not Assigned" - Show assignment date if available - Update status badges based on current inquiry status - User navigation and interaction: - Provide navigation back to home page - Show success/error messages from session - Handle empty state when no inquiries exist - Enable responsive design for mobile devices	

4.3.2 viewSubmittedInquiriesView [SDD-REQ-302]

Class Type	Boundary class	
Responsibility	An admin view interface that displays all submitted inquiries with search and filter functionality for assignment management.	
Attributes	Attributes Name	Attributes Type
	inquiries	collection
	InquiryID	integer
	InquiryTitle	string
	InquirySource	string
	InquiryStatus	string
	InquirySendDate	datetime
	search	string
	status	string

	agencies	collection
Methods	Method Name	Description
	route('admin.inquiries')	To display submitted inquiries list
	route('admin.inquiry.details/{id}')	To view specific inquiry details
	request('search')	To handle search functionality
	request('status')	To filter by status
Algorithm	1. Load submitted inquiries: - Fetch all inquiries with user relationships - Apply search filters on InquiryTitle, InquirySource, InquiryID - Apply status filter if provided - Sort by InquirySendDate desc with Pending first 2. Display inquiry management interface: - Show table with ID, Title, Source, Status, Date, Action columns - Display search and filter controls - Show "Review" button for each inquiry - Display status badges with color coding 3. Handle assignment preparation: - Identify unassigned inquiries (AgencyID is null) - Show assignment status for each inquiry - Provide navigation to assignment interface	

4.3.3 assignInquiryToPageView [SDD-REQ-303]

Class Type	Boundary class	
Responsibility	An admin view interface for assigning inquiries to agencies with bulk assignment capabilities and agency selection.	
Attributes	Attributes Name	Attributes Type
	inquiries	collection
	agencies	collection
	inquiry_ids	array
	agency_id	integer
	InquiryID	integer
	InquiryTitle	string
	InquiryStatus	string
	AgencyID	integer

	AgencyName	string
	selectedCount	integer
Methods	Method Name	Description
	route('admin.assign.inquiry')	To display assignment page
	route('admin.assign.inquiries')	To process bulk assignments
	showPage(pageId)	To navigate between assignment views
	updateSelectedCount()	To track selected inquiries
Algorithm	1. Display assignment interface: a. Load all inquiries with agency and user relationships b. Load all available agencies for assignment c. Show checkbox selection for multiple inquiries d. Display agency dropdown for assignment 2. Handle bulk assignment: a. Validate inquiry_ids array and agency_id b. Update selected inquiries with AgencyID c. Set assignment_date to current timestamp d. Redirect with success message 3. Assignment management: a. Show current assignment status for each inquiry b. Display "Select All" functionality with counter c. Provide agency selection dropdown d. Handle assignment validation and processing	

4.3.4 addAssignmentNotesAndInstructionsView [SDD-REQ-304]

Class Type	Boundary class	
Responsibility	An admin view interface for adding detailed assignment notes and instructions when assigning inquiries to agencies.	
Attributes	Attributes Name	Attributes Type
	inquiry_ids	array
	agency_id	integer
	assignment_notes	text
	priority_level	string
	expected_completion	date
	special_instructions	text
	selectedInquiries	collection
	targetAgency	object
Methods	Method Name	Description
	route('admin.assign.inquiries.with.notes')	To process assignment with notes

	validateAssignmentForm()	To validate assignment data
	showAssignmentPreview()	To preview assignment details
Algorithm	1. Display assignment notes interface: - Show selected inquiries for assignment - Display target agency information - Provide textarea for detailed assignment notes - Show priority level selection (Normal, High, Urgent) 2. Handle assignment with notes: - Validate assignment_notes and priority_level - Set expected_completion date if provided - Update inquiries with AgencyID and assignment details - Store assignment_notes and priority_level 3. Assignment tracking: - Set assignment_date to current timestamp - Update inquiry status to "Under Investigation" - Store special instructions for agency reference - Send assignment notifications to agencies	

4.3.5 generateAssignedInquiryReportsView [SDD-REQ-305]

Class Type	Boundary class	
Responsibility	An admin view interface for generating comprehensive reports on inquiry assignments with various filtering and export options.	
Attributes	Attributes Name	Attributes Type
	agencies	collection
	report_type	string
	date_from	date
	date_to	date
	agency_id	integer
	status	string
	format	string
	include_charts	string
	assignedInquiries	collection
	assignmentStats	array
Methods	Method Name	Description
	route('admin.reports')	To display assignment reports page
	route('admin.generate.assignment.reports')	To generate assignment reports
	generateAssignmentReports()	To process report creation

Algorithm	1. Assignment report configuration: a. Set date range for assignment filtering b. Select report type (Summary, Detailed, Statistics) c. Choose target agency or all agencies d. Apply status filters for assigned inquiries 2. Data collection and analysis: a. Filter assigned inquiries by date range and agency b. Calculate assignment statistics per agency c. Count inquiries by status: Pending, Under Investigation, etc. d. Generate assignment performance metrics 3. Report generation: a. Compile assignment data with agency details b. Calculate assignment completion rates c. Generate charts for assignment distribution d. Export in PDF or Excel format with assignment timelines 4. Assignment analytics: a. Track assignment patterns and trends b. Monitor agency workload distribution c. Calculate average investigation times d. Identify bottlenecks in assignment process
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4.3.6 displayAndApprovedView [SDD-REQ-306]

Class Type	Boundary class	
Responsibility	An agency view interface that displays all inquiries assigned to the agency with management capabilities including status updates, rejection, and approval	
Attributes	Attributes Name	Attributes Type
	agency	object
	inquiries	collection
	statusCounts	array
	InquiryID	integer
	InquiryTitle	string
	InquiryDescription	text
	InquiryNewsSource	string
	InquiryEvidence	string
	InquiryStatus	string
	InquiryPriority	string
	assignment_date	datetime
	user	object
	filterStatus	string

	filterPriority	string
Methods	Method Name	Description
	route('agency.view.display.inquiry')	To display assigned inquiries page
	showInquiryDetails(inquiryId)	To show detailed inquiry modal
	rejectInquiry(inquiryId)	To reject an inquiry with reason
	acceptInquiry(inquiryId)	To approve an inquiry with comments
	filterInquiries()	To filter inquiries by status/priority
	clearFilters()	To clear all applied filters
	asset('storage/' . evidence)	To generate evidence file URLs
Algorithm	- Load assigned inquiries: - Fetch inquiries where AgencyID matches current agency - Load with user, agency, and admin relationships - Order by priority: Urgent, High, Normal, then by assignment_date desc - Calculate status counts for dashboard display - Display inquiry management interface: - Show summary cards with total, under_investigation, verified_true, identified_fake counts - Display filter controls for status: All, Pending, Under Investigation, Verified as True, Identified as Fake - Display filter controls for priority: All Priorities, Urgent, High, Normal - Show inquiry cards with InquiryID, InquiryTitle, status badges, priority levels - Handle inquiry interactions: - Show detailed modal with complete inquiry information using AJAX - Display evidence files with download links using asset('storage/') - Show user contact information and submission details - Provide action buttons: "Show Full Details", "Reject Inquiry", "Accept Inquiry" - Status and priority management: - Display status badges with color coding: Pending (yellow), Under Investigation (blue), Verified as True (green), Identified as Fake (red) - Show priority badges with appropriate styling: Urgent (red), High (orange), Normal (gray) - Apply real-time filtering by status and priority using JavaScript - Update inquiry counts dynamically based on applied filters - Inquiry processing actions: - Handle acceptance with required comments validation - Process rejection with reason and comments requirements - Update InquiryStatus and StatusComments in database - Set ProcessedAt timestamp for tracking - Return JSON responses for AJAX compatibility - Evidence file handling: - Parse InquiryEvidence (JSON string or single file path) - Display evidence items with appropriate file type icons - Generate secure download links using asset('storage/' . file) - Open files in new tab for viewing without affecting current page	

4.3.7 InquiryAssignmentController [SDD-REQ-307]

Class Type	Control class	
Responsibility	A controller class that manages the assignment of inquiries to agencies with bulk assignment capabilities and assignment tracking.	
Attributes	Attributes Name	Attributes Type
	inquiry_ids	array
	agency_id	integer
	assignment_notes	text
	priority_level	string
	inquiries	collection
	agencies	collection
	assignmentData	array
Methods	Method Name	Description
	showAssignInquiry()	To display inquiry assignment interface
	assignInquiries()	To process bulk inquiry assignments
	assignInquiriesWithNotes()	To assign inquiries with detailed notes
	validateAssignmentData()	To validate assignment parameters
Algorithm	1. Display assignment interface: a. Load all inquiries with agency and user relationships b. Load all available agencies for assignment c. Show checkboxes for bulk selection d. Display agency dropdown and assignment controls 2. Process bulk assignments: a. Validate inquiry_ids array and agency_id b. Check that selected inquiries are not already assigned c. Update selected inquiries with AgencyID d. Set assignment_date to current timestamp e. Redirect with success message 3. Handle assignment with notes: a. Validate assignment_notes and priority_level b. Set expected_completion date if provided c. Update inquiries with AgencyID and assignment details d. Store assignment_notes and priority_level in database e. Send assignment notifications to agencies	

4.3.8 AdminController [SDD-REQ-308]

Class Type	Control class
Responsibility	A controller class that manages admin operations for Module 3 including inquiry review, assignment management, and report generation.

Attributes	Attributes Name	Attributes Type
	admin_id	integer
	admin_name	string
	inquiries	collection
	agencies	collection
	search	string
	status	string
	reportData	array
Methods	Method Name	Description
	showInquiries()	To display all submitted inquiries for review
	showInquiryDetails()	To show detailed inquiry information
	showAssignInquiry()	To display assignment page
	generateAssignmentReports()	To create assignment reports
Algorithm	1. Load inquiries for review: - Fetch all inquiries with user relationships - Apply search filters on InquiryTitle, InquirySource, InquiryID - Apply status filter if provided - Sort by InquirySendDate desc with Pending first 2. Display assignment management: - Load inquiries with agency and user relationships - Show current assignment status - Provide assignment controls and agency selection 3. Generate assignment reports: - Filter assigned inquiries by date range and agency - Calculate assignment statistics per agency - Generate assignment performance metrics - Export in PDF or Excel format with assignment timelines	

4.3.9 UserController [SDD-REQ-309]

Class Type	Control class	
Responsibility	A controller class that handles public user operations for Module 3 including viewing assigned agencies and tracking inquiry assignments.	
Attributes	Attributes Name	Attributes Type
	userSession	object
	userInquiries	collection
	InquiryID	integer
	AgencyID	integer

	assignmentStatus	string
	previous_url	string
Methods	Method Name	Description
	viewAssignedAgencies()	To show user's inquiries with assigned agencies
	showInquiryAssignmentStatus()	To display assignment tracking
	index()	To display user's inquiry list with assignments
Algorithm	1. Display assigned agencies: a. Fetch inquiries where UserID matches authenticated user b. Load with agency relationship to show assignment details c. Order by InquirySendDate desc d. Show assignment status and agency information 2. Track assignment status: a. Check if inquiry has been assigned to agency (AgencyID not null) b. Display agency name if assigned, otherwise show "Not Assigned" c. Show assignment date and current investigation status d. Provide estimated completion timeline	

4.3.10 AgencyController [SDD-REQ-310]

Class Type	Control class	
Responsibility	A controller class that manages agency operations for Module 3 including viewing assigned inquiries, processing assignments, and status updates.	
Attributes	Attributes Name	Attributes Type
	agency_id	integer
	agency_name	string
	assignedInquiries	collection
	statusCounts	array
	InquiryStatus	string
	priority_level	string
Methods	Method Name	Description
	showViewAndDisplayInquiry()	To display assigned inquiries dashboard
	showInquiryDetails()	To show specific inquiry details
	acceptInquiry()	To accept an assigned inquiry
	rejectInquiry()	To reject an assigned inquiry

Algorithm	1. Display assigned inquiries: a. Fetch inquiries where AgencyID matches current agency b. Load with user, agency, and admin relationships c. Order by priority: Urgent, High, Normal, then by assignment_date d. Calculate status counts for dashboard 2. Handle inquiry processing: a. Verify inquiry belongs to current agency b. For acceptance: validate comments and update status c. For rejection: validate reason and comments d. Update InquiryStatus and StatusComments e. Set ProcessedAt timestamp 3. Assignment status management: a. Display inquiries with priority levels b. Show status badges and investigation progress c. Handle AJAX requests for real-time updates
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4.3.11 InquiryController [SDD-REQ-311]

Class Type	Control class	
Responsibility	A controller class that manages inquiry operations specific to Module 3 assignment functionality including inquiry tracking and assignment status updates	
Attributes	Attributes Name	Attributes Type
	InquiryID	integer
	AgencyID	integer
	assignment_date	datetime
	assignmentHistory	array
	userInquiries	collection
	inquiry	object
Methods	Method Name	Description
	index()	To display user inquiries with assignment info
	show()	To display specific inquiry details
	updateAssignmentStatus()	To update inquiry assignment status
	trackAssignmentProgress()	To track assignment progress

Algorithm	1. Display inquiry list with assignments: a. Ensure user is authenticated b. Get user's inquiries with agency relationships c. Order by InquirySendDate desc d. Show assignment status and progress 2. Track assignment progress: a. Load inquiry with assignment history b. Check current assignment status c. Calculate investigation timeline d. Show progress indicators 3. Handle assignment updates: a. Update assignment_date when inquiry is assigned b. Track status changes throughout investigation c. Maintain assignment history log d. Send notifications on status updates
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4.3.12 PublicUsers [SDD-REQ-312]

Class Type	Entity class	
Responsibility	A model class that represents public users who submit inquiries and can view their assignment status to agencies.	
Attributes	Attributes Name	Attributes Type
	UserID	integer (Primary Key)
	UserName	string
	UserEmail	string
	UserPassword	string
	UserPhoneNum	string
	UserProfilePicture	string
	created_at	datetime
	updated_at	datetime
Methods	Method Name	Description
	inquiries()	To get all inquiries submitted by the user
	assignedInquiries()	To get inquiries with assigned agencies
	getAuthPassword()	To get password for authentication
	createUser()	To create a new public user
	updateProfile()	To update user profile information

Algorithm	1. Inquiry assignment tracking: a. Retrieve user's submitted inquiries with agency relationships b. Check assignment status for each inquiry c. Display agency information if assigned d. Show "Not Assigned" status for unassigned inquiries 2. User authentication for Module 3: a. Validate user credentials during login b. Maintain session for assignment viewing c. Ensure secure access to assignment status 3. Assignment status visibility: a. Show only user's own inquiry assignments b. Display agency names and assignment dates c. Track investigation progress
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4.3.13 Administrator [SDD-REQ-313]

Class Type	Entity class	
Responsibility	A model class that represents administrators who manage inquiry assignments to agencies with comprehensive assignment tracking and reporting capabilities.	
Attributes	Attributes Name	Attributes Type
	AdminID	integer (Primary Key)
	AdminName	string
	AdminEmail	string
	AdminUserName	string
	AdminPassword	string
	AdminRole	string
	AdminPhoneNum	string
	AdminAddress	string
	AdminProfilePicture	string
	created_at	datetime
	updated_at	datetime
Methods	Method Name	Description
	assignedInquiries()	To get all inquiries assigned by this admin
	bulkAssignInquiries()	To assign multiple inquiries to agencies
	generateAssignmentReports()	To create assignment reports
	checkPassword()	To verify admin password
	findByLogin()	To find admin by username or email

Algorithm	1. Bulk inquiry assignment: a. Validate inquiry_ids array and target agency b. Check that inquiries are not already assigned c. Update inquiries with AgencyID and assignment_date d. Store assignment notes and priority levels e. Send notifications to assigned agencies 2. Assignment tracking and management: a. Monitor assignment distribution across agencies b. Track assignment completion rates c. Generate performance metrics for agencies d. Maintain assignment history and audit trail 3. Assignment report generation: a. Filter assignments by date range and agency b. Calculate assignment statistics and trends c. Generate comprehensive assignment reports d. Export reports in PDF and Excel formats
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4.3.14 AssignedInquiry [SDD-REQ-314]

Class Type	Entity class	
Responsibility	A model class that represents the assignment relationship between inquiries and agencies with detailed assignment tracking information.	
Attributes	Attributes Name	Attributes Type
	AdminID	integer (Foreign Key)
	AgencyID	integer (Foreign Key)
	InquiryID	integer (Foreign Key)
	AssignedDate	date
	AdminNote	string
	assignment_notes	text
	priority_level	string
	expected_completion	date
Methods	Method Name	Description
	inquiry()	To get the assigned inquiry details
	agency()	To get the assigned agency information
	administrator()	To get the admin who made the assignment
	createAssignment()	To create new assignment record
	updateAssignmentStatus()	To update assignment progress

Algorithm	1. Assignment record creation: a. Validate AdminID, AgencyID, and InquiryID b. Set AssignedDate to current date c. Store AdminNote and assignment details d. Set priority_level (Normal, High, Urgent) e. Calculate expected_completion date 2. Assignment relationship management: a. Link inquiries to agencies through assignment b. Track admin who made the assignment c. Maintain assignment history and changes d. Handle assignment transfers between agencies 3. Assignment status tracking: a. Monitor assignment progress and completion b. Track time from assignment to resolution c. Generate assignment performance metrics d. Alert on overdue assignments
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4.3.15 Agency [SDD-REQ-315]

Class Type	Entity class	
Responsibility	A model class that represents agencies who receive assigned inquiries and process them with acceptance, rejection, and status update capabilities.	
Attributes	Attributes Name	Attributes Type
	AgencyID	integer (Primary Key)
	AgencyName	string
	AgencyUserName	string
	AgencyEmail	string
	AgencyPassword	string
	AgencyPhoneNum	string
	AgencyType	string
	created_at	datetime
	updated_at	datetime
Methods	Method Name	Description
	assignedInquiries()	To get all inquiries assigned to this agency
	acceptInquiry()	To accept an assigned inquiry
	rejectInquiry()	To reject an assigned inquiry
	updateInquiryStatus()	To update inquiry investigation status
	checkPassword()	To verify agency password
	findByLogin()	To find agency by username or email

Algorithm	1. Assigned inquiry management: a. Retrieve inquiries where AgencyID matches current agency b. Order by priority: Urgent, High, Normal, then by assignment_date c. Calculate status counts for dashboard display d. Filter inquiries by status and priority 2. Inquiry processing workflow: a. Accept inquiry: validate comments and update status b. Reject inquiry: require reason and detailed comments c. Update InquiryStatus and StatusComments d. Set ProcessedAt timestamp for tracking e. Send status notifications to users and admins 3. Agency authentication and security: a. Verify agency credentials during login b. Ensure access only to assigned inquiries c. Maintain secure session management d. Handle password updates and security settings
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4.3.16 Inquiry [SDD-REQ-316]

Class Type	Entity class	
Responsibility	A model class that represents inquiries with assignment functionality, tracking their assignment status, assigned agencies, and processing progress.	
Attributes	Attributes Name	Attributes Type
	InquiryID	integer (Primary Key)
	InquiryTitle	string
	InquiryDescription	text
	InquiryNewsSource	string
	InquiryEvidence	string
	InquiryStatus	string
	InquiryPriority	string
	VerificationDescription	text
	InquirySendDate	datetime
	AgencyID	integer (Foreign Key)
	assignment_date	datetime
	AdminID	integer (Foreign Key)
	UserID	integer (Foreign Key)
	StatusComments	text
	ProcessedAt	datetime
Methods	Method Name	Description
	agency()	To get the assigned agency information

	user()	To get the user who submitted the inquiry
	admin()	To get the admin who assigned the inquiry
	assignedInquiry()	To get assignment details
	updateAssignmentStatus()	To update assignment and processing status
	isAssigned()	To check if inquiry is assigned to an agency
Algorithm	1. Assignment status management: a. Check if AgencyID is not null to determine assignment b. Track assignment_date for timeline calculation c. Show "Not Assigned" or agency name in user views d. Update assignment status when agencies process inquiries 2. Inquiry processing workflow: a. Track inquiry from submission to assignment b. Monitor agency processing: acceptance, rejection, investigation c. Update InquiryStatus based on agency actions d. Store StatusComments from agency processing e. Set ProcessedAt timestamp when completed 3. Assignment relationships: a. Link to assigned agency through AgencyID b. Track admin who made the assignment through AdminID c. Maintain user relationship through UserID d. Support assignment history and transfers 4. Priority and status tracking: a. Handle InquiryPriority levels: Normal, High, Urgent b. Track status progression: Pending → Under Investigation → Verified/Rejected c. Calculate investigation timelines and performance metrics d. Generate assignment and processing reports	

4.3.17 assignInquirySectionView [SDD-REQ-317]

Class Type	Boundary class	
Responsibility	An admin view interface that provides a comprehensive inquiry assignment dashboard with multiple assignment workflows, statistics, and bulk management capabilities.	
Attributes	Attributes Name	Attributes Type
	inquiries	collection
	agencies	collection
	inquiry_ids	array
	agency_id	integer
	assignment_notes	text
	priority_level	string
	expected_completion	date

	selectedCount	integer
	statsData	object
	search	string
	status	string
Methods	Method Name	Description
	route('admin.assign.inquiry')	To display assign inquiry section
	route('admin.assign.inquiries')	To process bulk assignments
	route('admin.assign.inquiries.with.notes')	To assign with detailed notes
	showPage(pageId)	To navigate between assignment workflows
	updateSelectedCount()	To track selected inquiries
	validateAssignmentForm()	To validate assignment data

Algorithm	1. Display assignment section dashboard: a. Show icon-based navigation with 4 main options: - Assign Inquiry: Basic bulk assignment functionality - View Submitted: Review all submitted inquiries - Assignment Notes: Assign with detailed instructions - Generate Reports: Create assignment reports b. Load statistics: Total, Unassigned, Pending Review, Available Agencies c. Provide seamless navigation between workflows 2. Bulk assignment workflow: a. Display inquiries table with checkbox selection b. Show "Select All" functionality with live counter c. Provide agency dropdown for assignment target d. Process bulk assignments: validate inquiry_ids and agency_id e. Update selected inquiries with AgencyID and assignment_date 3. Assignment with notes workflow: a. Extended assignment form with detailed instructions b. Collect assignment_notes (required text area) c. Set priority_level options: Normal, High, Urgent d. Optional expected_completion date selection e. Validate and store comprehensive assignment data 4. View submitted inquiries workflow: a. Display all inquiries with current assignment status b. Show assigned agency or "Not Assigned" status c. Filter by status, agency, and search functionality d. Provide "Review" action buttons for detailed view 5. Interactive features: a. Real-time selected count updates using JavaScript b. Dynamic page switching with showPage() function c. Form validation before submission d. Success/error message display 6. Filter and search capabilities: a. Search by InquiryTitle, InquirySource, or InquiryID b. Filter by status: Pending, Under Investigation, Verified as True, Identified as Fake, Rejected c. Filter by agency: All Agencies, Unassigned, specific agencies d. Clear filters functionality 7. Assignment tracking and management: a. Display assignment history and notes b. Show priority levels with color-coded badges c. Track assignment completion dates d. Monitor agency workload distribution
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4.2 Module 4: Inquiry Progress Tracking [SDD-REQ-400]

4.3.1 View Inquiry Status Page [SDD-REQ-401]

Class Type	View	
Responsibility	Display detailed status and progress information for a specific inquiry	
Attributes	Attributes Name	Attributes Type
	inquiry_id	Integer
	inquiry_status	String
	progress_percentage	Float
	status_history	Array
	last_updated	DateTime
Methods	Method Name	Description
	show(\$id)	Display specific inquiry details and status (InquiryController)
	getInquiryProgress(\$inquiryId)	Get comprehensive progress data (ProgressTrackingController)
Algorithm	Status tracking algorithm with progress calculation and timeline visualization	

4.3.2 Inquiry Dashboard Page [SDD-REQ-402]

Class Type	View	
Responsibility	Provide overview dashboard of all inquiries with status summaries	
Attributes	Attributes Name	Attributes Type
	user_type	String
	status_counts	Array
	recent_inquiries	Array
	dashboard_stats	Object
Methods Algorithm	Method Name	Description
	index()	Display inquiry dashboard with filtering (InquiryController)
	getProgressDashboard()	Get dashboard for different user types (ProgressTrackingController)
	getDashboardStats()	Get dashboard statistics (ReportController)

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4.3.2 Filter Inquiry Page [SDD-REQ-403]

Class Type	View	
Responsibility	Allow users to filter and search inquiries by various criteria	
Attributes	Attributes Name	Attributes Type
	filter_criteria	Array
	search_term	String
	date_range	Object
	status_filter	String
Methods	Method Name	Description
	filter(Request \$request)	Filter inquiries by various criteria (InquiryController)
	index(Request \$request)	Display filtered inquiry list (InquiryController)
Algorithm	Advanced filtering algorithm with multiple criteria support and search optimization	

4.3.2 Notification Page [SDD-REQ-404]

Class Type	View	
Responsibility	Display notifications to users based on their type and permissions	
Attributes	Attributes Name	Attributes Type
	user_type	String
	notifications	Array
	unread_count	Integer
	notification_filters	Object
Methods	Method Name	Description
	publicNotifications()	Display notifications for public users (NotificationController)
	adminNotifications()	Display notifications for admin (NotificationController)
	filter(Request \$request)	Filter notifications by criteria (NotificationController)
	markAsRead(\$notificationId)	Mark notification as read (NotificationController)
Algorithm	Multi-user notification algorithm with real-time updates and filtering	

4.3.2 Monitor Agency Progress Page [SDD-REQ-405]

Class Type	View	
Responsibility	Monitor progress and performance of agencies	
Attributes	Attributes Name	Attributes Type
	agency_id	Integer
	performance_metrics	Object
	workload_data	Array
	progress_charts	Object
Methods	Method Name	Description
	showHome()	Show agency home dashboard (AgencyController)
	getAgencyPerformance()	Get agency performance metrics (ProgressTrackingController)
	getWorkloadDistribution()	Get workload distribution (ProgressTrackingController)
Algorithm	Agency performance tracking algorithm with workload analysis and KPI calculations	

4.3.2 View Agency Notes Page [SDD-REQ-406]

Class Type	View	
Responsibility	Display investigation notes created by agencies	
Attributes	Attributes Name	Attributes Type
	inquiry_id	Integer
	agency_notes	Array
	attachments	Array
	note_history	Array
Methods	Method Name	Description
	show(\$inquiryId)	Display investigation notes for admin (InvestigationNoteController)
	agencyView(\$inquiryId)	Display notes for agency view (InvestigationNoteController)
	getNotesForInquiry(\$inquiryId)	API endpoint for getting notes (InvestigationNoteController)
Algorithm	Note management algorithm with version control and access permissions	

4.3.2 Generate Agency Report Page [SDD-REQ-407]

Class Type	View	
Responsibility	Generate comprehensive reports for agency performance	
Attributes	Attributes Name	Attributes Type
	report_type	String
	date_range	Object
	agency_filter	Integer
	export_format	String
Methods	Method Name	Description
	generateAgencyReport(Request \$request)	Generate agency performance reports (ReportController)
	exportToPDF(Request \$request)	Export reports to PDF (ReportController)
	exportToExcel(Request \$request)	Export reports to Excel (ReportController)

Algorithm	Report generation algorithm with multiple formats and customizable parameters
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4.3.2 Report Dashboard Page [SDD-REQ-408]

Class Type	View	
Responsibility	Main dashboard for all reporting features and analytics	
Attributes	Attributes Name	Attributes Type
	report_categories	Array
	visual_statistics	Object
	trend_data	Array
	dashboard_widgets	Array
Methods	Method Name	Description
	index()	Display reports dashboard (ReportController)
	getVisualStatistics()	Get data for charts and graphs (ReportController)
	generateTrendAnalysis()	Generate monthly trend analysis (ReportController)
Algorithm	Analytics dashboard algorithm with real-time data visualization and trend analysis	

4.3.2 View Assigned Inquiries Page [SDD-REQ-409]

Class Type	View	
Responsibility	Display inquiries assigned to current user	
Attributes	Attributes Name	Attributes Type
	user_id	Integer
	assigned_inquiries	Array
	inquiry_status	String
	assignment_date	DateTime
Methods	Method Name	Description
	showViewAndDisplayInquiry()	Show agency view and display inquiry page (AgencyController)
	index(Request \$request)	Display filtered inquiry list (InquiryController)

Algorithm	Assignment tracking algorithm with user-specific filtering and status monitoring
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4.3.2 Update Inquiry Details Page [SDD-REQ-410]

Class Type	View	
Responsibility	Allow users to update inquiry information and status	
Attributes	Attributes Name	Attributes Type
	inquiry_id	Integer
	editable_fields	Array
	status_options	Array
	update_permissions	Object
Methods	Method Name	Description
	edit(\$id)	Show form for editing inquiry (InquiryController)
	update(Request \$request, \$id)	Update existing inquiry (InquiryController)
	updateStatus(Request \$request, \$id)	Update inquiry status (InquiryController)
Algorithm	Update validation algorithm with permission checks and data integrity maintenance	

4.3.2 Add Investigation Notes Page [SDD-REQ-411]

Class Type	View	
Responsibility	Interface for adding investigation notes and attachments	
Attributes	Attributes Name	Attributes Type
	inquiry_id	Integer
	note_content	Text
	attachments	FileArray
	officer_details	Object
Methods	Method Name	Description
	create(\$inquiryId)	Show form for creating investigation note (InvestigationNoteController)
	store(Request \$request, \$inquiryId)	Store new investigation note (InvestigationNoteController)
	edit(\$noteId)	Show form to edit investigation note (InvestigationNoteController)
Algorithm	Note creation algorithm with file handling and notification triggers	

4.3.2 Request Clarification Page [SDD-REQ-412]

Class Type	View	
Responsibility	Allow agencies to request clarification from admin	
Attributes	Attributes Name	Attributes Type
	inquiry_id	Integer
	clarification_request	Text
	request_urgency	String
	contact_details	Object
Methods	Method Name	Description
	requestClarification(Request \$request)	Request clarification from admin (NotificationController)
	requestReassignment(Request \$request)	Request inquiry reassignment (NotificationController)
Algorithm	Communication algorithm with priority handling and automatic routing	

4.3.2 InquiryController [SDD-REQ-413]

Class Type	Controller	
Responsibility	Handle all inquiry-related operations and business logic	
Attributes	Attributes Name	Attributes Type
	inquiry_model	Object
	validation_rules	Array
	authorization_rules	Array
Methods	Method Name	Description
	store(Request \$request)	Store new inquiry
	index(Request \$request)	Display inquiry list with filtering
	show(\$id)	Display specific inquiry details
	create()	Show form for creating new inquiry
	edit(\$id)	Show form for editing inquiry

	update(Request \$request, \$id)	Update existing inquiry
	destroy(\$id)	Delete inquiry
	updateStatus(Request \$request, \$id)	Update inquiry status
	filter(Request \$request)	Filter inquiries by various criteria
Algorithm	CRUD operations algorithm with status management and authorization checks	

4.3.2 ReportController [SDD-REQ-414]

Class Type	Controller	
Responsibility	Generate reports and handle analytics operations	
Attributes	Attributes Name	Attributes Type
	report_generators	Array
	export_formats	Array
	analytics_data	Object
Methods	Method Name	Description
	index()	Display reports dashboard
	generateAgencyReport(Request \$request)	Generate agency performance reports
	exportToPDF(Request \$request)	Export reports to PDF
	exportToExcel(Request \$request)	Export reports to Excel
	getVisualStatistics(Request \$request)	Get data for charts and graphs
	generateStatusReport(Request \$request)	Generate comprehensive status reports
	generateTrendAnalysis(Request \$request)	Generate monthly trend analysis
	getDashboardStats()	Get dashboard statistics
Algorithm	Report generation algorithm with multiple formats and statistical analysis	

4.3.2 NotificationController [SDD-REQ-415]

Class Type	Controller
Responsibility	Manage all notification operations and communications

Attributes	Attributes Name	Attributes Type
	notification_types	Array
	recipient_rules	Object
	delivery_methods	Array
Methods	Method Name	Description
	publicNotifications()	Display notifications for public users
	adminNotifications()	Display notifications for admin
	sendStatusUpdateToPublic(Request \$request)	Send status updates to public users
	notifyInquiryCompleted(Request \$request)	Notify admin of inquiry completion
	requestReassignment(Request \$request)	Request inquiry reassignment
	requestClarification(Request \$request)	Request clarification from admin
	getUnreadCount()	Get unread notification count
	markAsRead(\$notificationId)	Mark notification as read
	filter(Request \$request)	Filter notifications by criteria
Algorithm	Multi-user notification algorithm with delivery optimization and read tracking	

4.3.2 Inquiry (Model) [SDD-REQ-416]

Class Type	Model	
Responsibility	Core inquiry data management and relationships	
Attributes	Attributes Name	Attributes Type
	InquiryID	Integer (Primary Key)
	InquiryTitle	String
	InquiryStatus	String
	UserID	Integer (Foreign Key)
	AgencyID	Integer (Foreign Key)
	InquirySendDate	DateTime
Methods	Method Name	Description
	user()	Relationship to PublicUser model
	agency()	Relationship to Agency model
	statusHistory()	Relationship to StatusHistory model
	investigationNotes()	Relationship to InvestigationNote model

Algorithm	Data integrity algorithm with relationship management and status validation
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4.3.2 Administrator (Model) [SDD-REQ-417]

Class Type	Model	
Responsibility	Administrator data management and authentication	
Attributes	Attributes Name	Attributes Type
	AdminID	Integer (Primary Key)
	AdminName	String
	AdminEmail	String (Unique)
	AdminPassword	String (Hashed)
Methods	Method Name	Description
	findByLogin(\$loginInput)	Find admin by login credentials
	checkPassword(\$password)	Verify admin password
Algorithm	Authentication algorithm with secure password handling and session management	

4.3.2 Agency (Model) [SDD-REQ-418]

Class Type	Model	
Responsibility	Agency data management and inquiry assignments	
Attributes	Attributes Name	Attributes Type
	AgencyID	Integer (Primary Key)
	AgencyName	String
	AgencyUserName	String (Unique)
	AgencyEmail	String (Unique)
	AgencyPassword	String (Hashed)
Methods	Method Name	Description
	inquiries()	Relationship to Inquiry model

	investigationNotes()	Relationship to InvestigationNote model
	findByLogin(\$loginInput)	Find agency by login credentials
	checkPassword(\$password)	Verify agency password
Algorithm	Agency management algorithm with workload tracking and performance metrics	

4.3.2 PublicUser (Model) [SDD-REQ-419]

Class Type	Model	
Responsibility	Public user data management and inquiry submissions	
Attributes	Attributes Name	Attributes Type
	Attributes Name	Attributes Type
	UserID	Integer (Primary Key)
	UserName	String
	UserEmail	String (Unique)
	UserPhoneNum	String
	UserPassword	String (Hashed)
Methods	Method Name	Description
	inquiries()	Relationship to Inquiry model
	notifications()	Relationship to Notification model
Algorithm	User management algorithm with inquiry tracking and notification preferences	

1. REQUIREMENT TRACEABILITY

5.1 Module 1: Manage User [SRS-REQ-100]

Requirement ID	Description	Design ID
SRS-REQ-101	Public users must register to access the system.	SDD-REQ-101, SDD-REQ-104, SDD-REQ-107
SRS-REQ-102	Registration requires a valid email address and a secure password.	SDD-REQ-101, SDD-REQ-107
SRS-REQ-103	Users must verify their email before gaining full access.	SDD-REQ-101, SDD-REQ-104
SRS-REQ-104	Registered users must log in using their email and password.	SDD-REQ-101, SDD-REQ-104, SDD-REQ-114
SRS-REQ-105	The system must support password recovery via email.	SDD-REQ-101, SDD-REQ-104, SDD-REQ-109, SDD-REQ-115
SRS-REQ-106	Users can view and edit their personal profile.	SDD-REQ-101, SDD-REQ-104, SDD-REQ-108
SRS-REQ-107	Editable fields include name, contact details, and profile picture.	SDD-REQ-101, SDD-REQ-108
SRS-REQ-108	Users can update their password and security settings.	SDD-REQ-101, SDD-REQ-104, SDD-REQ-109
SRS-REQ-109	MCMC staff must log in to access the system.	SDD-REQ-103, SDD-REQ-105, SDD-REQ-114
SRS-REQ-110	Login requires a valid staff username and password.	SDD-REQ-103, SDD-REQ-105, SDD-REQ-114
SRS-REQ-111	The system must support password recovery via email for MCMC staff.	SDD-REQ-103, SDD-REQ-105, SDD-REQ-115
SRS-REQ-112	MCMC staff are responsible for registering agencies involved in the system.	SDD-REQ-102, SDD-REQ-105, SDD-REQ-110
SRS-REQ-113	Each registered agency will be assigned a unique username and password.	SDD-REQ-102, SDD-REQ-105, SDD-REQ-110
SRS-REQ-114	Login credentials for agencies will be sent via email upon registration.	SDD-REQ-102, SDD-REQ-105, SDD-REQ-110
SRS-REQ-115	MCMC staff can view all registered user data within the system.	SDD-REQ-103, SDD-REQ-105, SDD-REQ-111
SRS-REQ-116	Access includes user profiles, registration details, and activity logs.	SDD-REQ-103, SDD-REQ-105, SDD-REQ-111
SRS-REQ-117	MCMC staff can generate reports on the number of registered users.	SDD-REQ-103, SDD-REQ-105, SDD-REQ-111
SRS-REQ-118	Reports can be filtered by date, user type, and agency.	SDD-REQ-103SDD-REQ-105SDD-REQ-111

SRS-REQ-119	The system must provide both summary and detailed reports in downloadable formats (e.g., PDF, Excel).	SDD-REQ-103, SDD-REQ-105, SDD-REQ-111
SRS-REQ-120	Agency staff can log in using the username and temporary password provided via email by MCMC.	SDD-REQ-102, SDD-REQ-106, SDD-REQ-114
SRS-REQ-121	Upon first login, agency staff must change their password to a new secure password.	SDD-REQ-102, SDD-REQ-106, SDD-REQ-114, SDD-REQ-117
SRS-REQ-122	The system must enforce strong password requirements.	SDD-REQ-101, SDD-REQ-102, SDD-REQ-103, SDD-REQ-115, SDD-REQ-117
SRS-REQ-123	Agency staff can update and manage their profile information.	SDD-REQ-102, SDD-REQ-106, SDD-REQ-112, SDD-REQ-116
SRS-REQ-124	Editable fields include agency name, contact details, and profile picture.	SDD-REQ-102, SDD-REQ-116
SRS-REQ-125	Agency staff can also update their password and security settings.	SDD-REQ-102, SDD-REQ-106, SDD-REQ-117

5.2 Module 2: Manage Inquiry Form Submission [SRS-REQ-200]

Requirement ID	Description	Design ID
SRS-REQ-201	Public users can submit inquiries to verify the authenticity of news.	SDD-REQ-201, SDD-REQ-211, SDD-REQ-217
SRS-REQ-202	Each inquiry must include supporting evidence (e.g., documents, images, or links).	SDD-REQ-201, SDD-REQ-206, SDD-REQ-211
SRS-REQ-203	Users must provide detailed information regarding the news in question.	SDD-REQ-201, SDD-REQ-206, SDD-REQ-211
SRS-REQ-204	Users can view a list of their previously submitted inquiries.	SDD-REQ-202, SDD-REQ-206, SDD-REQ-217
SRS-REQ-205	Each inquiry record includes submission details and status updates.	SDD-REQ-202, SDD-REQ-203, SDD-REQ-206
SRS-REQ-206	Users can view all inquiries submitted by other individuals.	SDD-REQ-202, SDD-REQ-206, SDD-REQ-217
SRS-REQ-207	The system must ensure that sensitive personal information is not disclosed.	SDD-REQ-202, SDD-REQ-206, SDD-REQ-207
SRS-REQ-208	Inquiries should be presented in a structured format with search and filter options.	SDD-REQ-202, SDD-REQ-213, SDD-REQ-215
SRS-REQ-209	MCMC staff can access a list of newly submitted inquiries from public users.	SDD-REQ-212, SDD-REQ-213, SDD-REQ-205
SRS-REQ-210	Each inquiry includes the submitted details, supporting evidence, and user information.	SDD-REQ-212, SDD-REQ-213, SDD-REQ-206
SRS-REQ-211	MCMC staff must review and filter inquiries before forwarding them to the relevant agency.	SDD-REQ-212, SDD-REQ-204, SDD-REQ-205
SRS-REQ-212	They will assess whether the inquiry is a genuine request or a non-serious submission.	SDD-REQ-212, SDD-REQ-205, SDD-REQ-206
SRS-REQ-213	Non-serious inquiries can be flagged or discarded with an appropriate status update.	SDD-REQ-212, SDD-REQ-205, SDD-REQ-206
SRS-REQ-214	MCMC staff can view a record of all previously filtered inquiries.	SDD-REQ-213, SDD-REQ-205, SDD-REQ-206
SRS-REQ-215	The system should allow filtering and searching of past inquiries by date, status, and agency.	SDD-REQ-213, SDD-REQ-205, SDD-REQ-210
SRS-REQ-216	A secure audit log should be maintained for tracking actions taken on each inquiry.	SDD-REQ-206, SDD-REQ-210, SDD-REQ-205
SRS-REQ-217	MCMC staff can generate reports on the total number of inquiries received from public users.	SDD-REQ-214, SDD-REQ-205, SDD-REQ-206
SRS-REQ-218	Reports can be filtered and categorised based on various criteria, including monthly or yearly inquiry statistics.	SDD-REQ-214, SDD-REQ-205, SDD-REQ-206
SRS-REQ-219	Reports should be available in multiple formats, such as PDF and Excel.	SDD-REQ-214, SDD-REQ-205
SRS-REQ-220	The system should provide graphical representations (e.g., charts and graphs) for easy data analysis.	SDD-REQ-214, SDD-REQ-205
SRS-REQ-221	Reports must be exportable and downloadable for further review.	SDD-REQ-214, SDD-REQ-205
SRS-REQ-222	Agencies can only access and record of all past inquiries assigned to them.	SDD-REQ-215, SDD-REQ-208, SDD-REQ-218
SRS-REQ-223	The system should allow filtering and searching based on inquiry status (e.g., under investigation, verified as true, identified as fake, rejected).	SDD-REQ-215, SDD-REQ-216, SDD-REQ-218
SRS-REQ-224	The system should allow filtering and searching based on date range (e.g., by month or year).	SDD-REQ-215, SDD-REQ-216, SDD-REQ-218
SRS-REQ-225	The system should allow filtering and searching based on inquiry category or subject matter.	SDD-REQ-215, SDD-REQ-216, SDD-REQ-206

SRS-REQ-226	Agencies should be able to review the full history of each inquiry, including status changes, previous updates, and any communications with MCMC.	SDD-REQ-216, SDD-REQ-206, SDD-REQ-210
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5.3 Module 3: Inquiry Assignment [SRS-REQ-300]

Requirement ID	Description	Design ID
SRS-REQ-301	Users can see which agency their inquiry has been forwarded to for verification.	SDD-REQ-301, SDD-REQ-309, SDD-REQ-316
SRS-REQ-302	The system should display the agency name and the date the inquiry was assigned.	SDD-REQ-301, SDD-REQ-312, SDD-REQ-316
SRS-REQ-303	MCMC staff will review public user inquiries and assign them to the relevant agency for verification.	SDD-REQ-302, SDD-REQ-303, SDD-REQ-307, SDD-REQ-308
SRS-REQ-304	The system should provide a list of registered agencies for easy selection.	SDD-REQ-303, SDD-REQ-317, SDD-REQ-315
SRS-REQ-305	Staff can add comments or notes before forwarding the inquiry to the agency.	SDD-REQ-304, SDD-REQ-307, SDD-REQ-314
SRS-REQ-306	MCMC staff can generate reports on the number of inquiries assigned to each agency.	SDD-REQ-305, SDD-REQ-308, SDD-REQ-313
SRS-REQ-307	Reports can be filtered by date range (e.g., by month, year) and agency name.	SDD-REQ-305, SDD-REQ-308, SDD-REQ-313
SRS-REQ-308	The system should provide visual analytics (e.g., charts, graphs) to identify trends in inquiry distribution.	SDD-REQ-305, SDD-REQ-308
SRS-REQ-309	Reports should be available for download in formats such as PDF and Excel.	SDD-REQ-305, SDD-REQ-308
SRS-REQ-310	Agencies will receive inquiries assigned by MCMC for verification.	SDD-REQ-306, SDD-REQ-310, SDD-REQ-315
SRS-REQ-311	Each inquiry will include relevant details, supporting evidence, and submission history.	SDD-REQ-306, SDD-REQ-310, SDD-REQ-316
SRS-REQ-312	Agencies must review the inquiry to determine if it falls within their jurisdiction.	SDD-REQ-306, SDD-REQ-310, SDD-REQ-315
SRS-REQ-313	If the inquiry is within their scope, they will proceed with the verification process.	SDD-REQ-306, SDD-REQ-310, SDD-REQ-315
SRS-REQ-314	If the inquiry is outside their jurisdiction, they must notify MCMC and reject the assignment.	SDD-REQ-306, SDD-REQ-310, SDD-REQ-315
SRS-REQ-315	Agencies can reject an inquiry by providing a reason and notifying MCMC.	SDD-REQ-306, SDD-REQ-310, SDD-REQ-315
SRS-REQ-316	The system should allow agencies to add comments explaining why the inquiry does not fall under their authority.	SDD-REQ-306, SDD-REQ-310, SDD-REQ-316
SRS-REQ-317	MCMC will then reassign the inquiry to a more appropriate agency or take further action.	SDD-REQ-307, SDD-REQ-308, SDD-REQ-313, SDD-REQ-314

5.4 Module 4: Inquiry Progress Tracking [SRS-REQ-400]

Requirement ID	Description	Design ID
SRS-REQ-401	Public users can track the status of their submitted inquiries in real-time.	SDD-REQ-401, SDD-REQ-402, SDD-REQ-405
SRS-REQ-402	The system should display whether an inquiry is Under Investigation, Verified as True, Identified as Fake, or Rejected.	SDD-REQ-401, SDD-REQ-402
SRS-REQ-403	Users will receive notifications when there is a status update on their inquiry.	SDD-REQ-404, SDD-REQ-415
SRS-REQ-404	The system should provide a detailed history log of status changes, including timestamps and agency actions.	SDD-REQ-401, SDD-REQ-413
SRS-REQ-405	A dedicated section in the user dashboard should display all inquiries with their current statuses.	SDD-REQ-402, SDD-REQ-405
SRS-REQ-406	The system should allow users to filter and search their past inquiries.	SDD-REQ-403, SDD-REQ-413
SRS-REQ-407	MCMC staff can track the progress of inquiries assigned to agencies.	SDD-REQ-405, SDD-REQ-413
SRS-REQ-408	The system should display real-time updates on each inquiry's status, including Under Investigation, Verified as True, Identified as Fake, or Rejected.	SDD-REQ-401, SDD-REQ-402, SDD-REQ-405
SRS-REQ-409	Agencies can provide progress notes and updates on their investigation.	SDD-REQ-406, SDD-REQ-411, SDD-REQ-413
SRS-REQ-410	MCMC staff can generate reports on agency performance, including number of inquiries assigned to each agency, number resolved, and average time taken to resolve an inquiry.	SDD-REQ-407, SDD-REQ-408, SDD-REQ-414
SRS-REQ-411	Reports should be filterable by date range, agency, and inquiry category.	SDD-REQ-407, SDD-REQ-408, SDD-REQ-414
SRS-REQ-412	The system should provide visual analytics such as charts and graphs to highlight trends and performance metrics.	SDD-REQ-408, SDD-REQ-414
SRS-REQ-413	Reports should be available in multiple formats, including PDF and Excel.	SDD-REQ-407, SDD-REQ-408, SDD-REQ-414
SRS-REQ-414	Agencies must update the status of each inquiry assigned by MCMC.	SDD-REQ-409, SDD-REQ-410, SDD-REQ-413
SRS-REQ-415	The system should allow agencies to set the following statuses: Under Investigation, Verified as True, Identified as Fake, or Rejected.	SDD-REQ-409, SDD-REQ-410, SDD-REQ-413
SRS-REQ-416	Agencies can add investigation notes, supporting documents, and explanations for their conclusions.	SDD-REQ-411, SDD-REQ-413, SDD-REQ-418
SRS-REQ-417	Each status update should include a timestamp and the name of the reviewing officer.	SDD-REQ-410, SDD-REQ-411, SDD-REQ-413
SRS-REQ-418	Agencies can notify MCMC when an inquiry is completed or if further clarification is needed.	SDD-REQ-412, SDD-REQ-415, SDD-REQ-418
SRS-REQ-419	The system should allow agencies to request reassignment if an inquiry does not fall under their authority.	SDD-REQ-412, SDD-REQ-415, SDD-REQ-418